

Therapy for Stage IV Non–Small Cell Lung Cancer With Driver Alterations: ASCO Living Guideline, 2026.3.0

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ABSTRACT

ASCO Guidelines provide recommendations with comprehensive review and analyses of the relevant literature for each recommendation, following the guideline development process as outlined in the *ASCO Guidelines Methodology Manual*. ASCO Guidelines follow the *ASCO Conflict of Interest Policy for Clinical Practice Guidelines*.

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PURPOSE To provide evidence-based recommendations for patients with stage IV non–small cell lung cancer with driver alterations.

METHODS This ASCO living guideline offers continually updated recommendations based on an ongoing systematic review of randomized controlled trials (RCTs), with the latest time frame spanning March–October 2025. An Expert Panel of medical oncology, pulmonary, community oncology, research methodology, and advocacy experts was convened. The literature search included systematic reviews, meta-analyses, and RCTs. Outcomes of interest include efficacy and safety. Expert Panel members used available evidence and informal consensus to develop evidence-based guideline recommendations.

RESULTS This guideline consolidates all previous updates and reflects the body of evidence informing this guideline topic. Thirteen studies were identified in the latest search of literature to date.

RECOMMENDATIONS Evidence-based recommendations were updated to address first, second, and subsequent treatment options for patients with driver alterations. Additional information is available at www.asco.org/thoracic-cancer-guidelines.

INTRODUCTION

The management of patients with metastatic non–small cell lung cancer (NSCLC) with driver alterations continues to evolve, with the emergence of treatment intensification strategies for selected subgroups and the regulatory approval of novel agents for others. This living guideline

continually incorporates the latest published research on systemic therapy in patients with driver alterations, including the management of special population patients such as those with poor performance status (PS). Since the last full version of the ASCO stage IV NSCLC with driver alterations living guideline published in 2024,¹ considerable progress has been made, with seven new regulatory approvals and

ACCOMPANYING CONTENT

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 Appendix

 Data Supplement

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TARGET POPULATION AND AUDIENCE

Target Population

Patients with stage IV NSCLC with targetable driver alterations.

Target Audience

Oncology care clinicians (including primary care physicians, specialists, nurses, social workers, and other relevant members of a comprehensive multidisciplinary cancer care team), patients, and their caregivers.

changes in first-line therapy for some of the driver alterations.

The approval of combination therapy for patients with classical *EGFR* mutations (del 19 and L858R) has raised new questions regarding the appropriate populations for treatment intensification. These and other key advances have clearly changed the management of patients with driver alterations and have prompted this full update.

The evidence base for any clinical practice recommendations is built on the characteristics of the patients who participated in the underlying studies. Broad enrollment in clinical trials is essential for clinicians to assess the generalizability of trial results and understand safety and efficacy across populations.

GUIDELINE QUESTIONS

This living clinical practice guideline addresses two overarching clinical questions: (1) What are the most effective first-line treatment options based on targetable driver alterations? (2) What are the most effective second-line and subsequent treatment options based on these driver alterations?

METHODS

Guideline Development Process

This systematic review-based guideline product was developed by a multidisciplinary Expert Panel, which included a representative from Ontario Health, a patient representative, and an ASCO guidelines staff member with health research methodology expertise (Appendix Table A1, online only).

The recommendations were developed using a systematic review of evidence identified through routinely scheduled searches of PubMed (the latest being from March 2025–October 2025) of phase III randomized controlled trials (RCTs). Articles were selected for inclusion in the systematic review based on the following criteria:

- Population: Patients with stage IV NSCLC whose test results show cancer harbors driver alterations in the following:
 - Epidermal growth factor receptor (*EGFR*) mutations, anaplastic lymphoma kinase (*ALK*) mutations, c-ROS oncogene 1 (*ROS1*) fusions, *BRAF*^{V600E} mutations, rearranged during transfection (*RET*) fusions, mesenchymal epithelial transition (*MET*) exon 14 skipping mutations, human epidermal growth factor receptor 2 (*HER2*) alterations, neurotrophic tyrosine receptor kinase (*NTRK*) fusions, and *KRAS* G12C mutations with known marker status test results available to the clinician.
- Interventions: Chemotherapy, monoclonal antibodies, targeted therapy, palliative care, no treatment
- Comparisons: Chemotherapy, monoclonal antibodies, targeted therapy, palliative care, no treatment
- Outcomes: Included progression-free survival (PFS), overall survival (OS), treatment toxicity (adverse events [AEs]; usually grade 3–4 AEs), overall response rates (ORR), and quality of life (if reported).
- Sample size: With driver alterations: Minimum sample size of 20 or 1% of target population required to have the actionable driver alteration of total participants.

Articles were excluded from the systematic review if they were (1) meeting abstracts not subsequently published in peer-reviewed journals; (2) editorials, commentaries, letters, news articles, case reports, narrative reviews; and (3) published in a non-English language.

For this latest iteration, one full panel meeting and three subgroup panel meetings were held. Members were asked to provide ongoing input on the guideline development protocol, quality and assessment of the evidence, generation of recommendations, and draft content. They also reviewed and approved drafts throughout the entire development of the guideline. ASCO staff met routinely with the Expert Panel co-chairs and corresponded with the Panel via e-mail to coordinate the process to completion. Ratings for the strength of the recommendation and evidence quality are provided with each recommendation, defined in Appendix Table A2. The quality of the evidence for each outcome was assessed using the Cochrane Risk of Bias tool and elements of the GRADE quality assessment and recommendations development process.^{2,3} GRADE quality assessment labels (ie, high, moderate, low, very low) were assigned for each outcome by the project methodologist in collaboration with the Expert Panel co-chairs and reviewed by the full Expert Panel. All funding for the administration of the project was provided by ASCO.

Guideline Review and Approval

The draft recommendations were released to the public for open comment from August 21 through September 3, 2025. Response categories of “Agree as written”, “Agree with suggested modifications,” and “Disagree. See comments” were captured for every proposed recommendation with 9

respondents and 13 written comments received. Of the 37 recommendations included in the open comment, only 5% of the responses disagreed with a recommendation, while the remaining 95% either agreed or agreed with slight modifications to the recommendations. Expert Panel members reviewed comments from all sources and determined whether to maintain original draft recommendations, revise with minor language changes, or consider major recommendation revisions.

All changes were incorporated into the final manuscript before ASCO Evidence Based Medicine Committee (EBMC) review and approval. All ASCO guidelines are ultimately reviewed and approved by the Expert Panel and the ASCO EBMC before submission to the *Journal of Clinical Oncology* for editorial review and consideration for publication.

Living Guideline Updating

The ASCO Expert Panel and guidelines staff will work with co-chairs to conduct a formal review of emerging literature every 2 weeks to assess the necessity and scope of updates. The ASCO Guidelines Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the living guideline update process. This is the most recent information as of the publication date.

Versioning and Numbering Statement

To ensure transparency and facilitate the tracking of evidence updates, ASCO has implemented a structured numbering system for living guidelines. This system reflects the status of the recommendations as follows:

- Major Updates (eg, 2026.3.0): These represent full updates or significant revisions approved by the Expert Panel and the Evidence Based Medicine Committee (EBMC). They typically incorporate comprehensive literature reviews and involves resubmission of the entire guideline.
- Rapid Updates (eg, 2026.3.1): These “mini updates” serve as timely notifications of revised recommendations or supporting evidence, ensuring the guideline remains current between full update cycles.

RESULTS

Characteristics of Studies Identified in the Literature Search

Of the 73 studies identified in the latest literature search that spanned from March 2025 to October 2025, 12 RCTs⁴⁻¹⁵ and one pooled analysis¹⁶ met the eligibility criteria. The randomized trials compared different therapies for driver alterations. Results from these studies either necessitated some modifications to the recommendations, or the quality of evidence rating for some recommendations. Discussions

of evidence supporting unchanged recommendations have been reviewed in the previous guideline publications.^{1,17-24}

RECOMMENDATIONS

Based on the body of evidence identified to date through several update iterations, complete recommendations for treating patients with stage IV NSCLC with driver alterations are provided in [Table 1](#).

BIOMARKER TESTING

Recommendation 1.0

Literature Review Update

The availability of biomarker testing results, ideally performed utilizing a broad next-generation sequencing (NGS) assay, in addition to immunohistochemistry (IHC) results for PD-L1, HER2, and MET, is of paramount importance to determine an appropriate treatment strategy, be that targeted therapy for NSCLC with an actionable driver mutation, or an immunotherapy-based strategy for NSCLC without an actionable alteration. Availability of comprehensive molecular profiling results prior to the initiation of frontline therapy has been associated with improved OS, as well as improved cost-effectiveness of care delivery.^{25,26} Combined molecular profiling utilizing both tissue and blood-based approaches can increase diagnostic yield and availability of informative results prior to initiation of therapy.^{25,27} In addition, utilizing a platform that incorporates RNA sequencing has the potential to improve identification of actionable gene fusions.^{27,28} Conversely, apart from resource-limited settings where there are no other options, single-gene polymerase chain reaction testing should not be routinely performed, given the decreased sensitivity of such an approach to identify actionable genomic alteration and overall inefficiency of tissue utilization.²⁹⁻³¹ Unfortunately, despite the established importance of molecular profiling in advanced NSCLC, many patients do not have complete molecular testing performed prior to initiation of therapy, with disparities noted according to patient race and medical insurance coverage.^{32,33}

Clinical Interpretation

The importance of obtaining comprehensive molecular profiling, ideally utilizing an NGS platform, combined with PD-L1 IHC testing, prior to initiation of first-line therapy, cannot be overstated. The results of said testing inform appropriate treatment decision making and improve survival. In the event that treatment must be started expediently to avoid imminent symptomatic or end-organ deterioration, initiation of histology-appropriate chemotherapy without immunotherapy can be pursued while awaiting molecular testing results.

FIRST-LINE THERAPY

EGFR

Exon 19 Deletion, Exon 21 L858R Substitution

Recommendation 1.1.

Literature review update. Version 2023.3 of the therapy for stage IV NSCLC with driver alterations living guideline recommended clinicians should offer osimertinib.¹ Osimertinib plus chemotherapy was added in version 2024.1,²¹ and lazertinib plus amivantamab in version 2024.2.²² This guideline update recommends clinicians offer osimertinib with platinum-pemetrexed chemotherapy or amivantamab plus lazertinib in first-line setting. Osimertinib remains an option for patients who decide not to pursue combination therapy (see recommendation 1.2). Over the past 2 years, two landmark studies have reported on the clinical benefits of combination therapy over osimertinib monotherapy in the first-line setting for patients with NSCLC whose tumors harbor a typical EGFR mutation.^{34,35} The first published study was FLAURA2, a phase III, international RCT, where 557 patients with previously untreated locally advanced or metastatic NSCLC with an EGFR mutation (exon 19 deletion or exon 21 L858R substitution) were randomly assigned 1:1 to receive osimertinib (80 mg once daily) plus platinum-based chemotherapy (cisplatin or carboplatin plus pemetrexed once every 3 weeks for four cycles, followed by maintenance pemetrexed; n = 279) or osimertinib monotherapy (80 mg once daily; n = 278). The primary end point of this study was PFS. The median PFS was significantly longer in the combination arm (25.5 v 16.7 months; hazard ratio [HR], 0.62 [95% CI, 0.49 to 0.79]; $P < .001$). ORR was similar between both treatment arms (83% v 76%, respectively).³⁴ OS also favored the combination (47.5 v 37.6 months; HR, 0.77 [95% CI, 0.61 to 0.96]; $P = .02$).¹¹ Notably, in a preplanned subgroup analysis, patients with CNS metastasis had significantly superior median PFS when treated with combination therapy (24.9 months v 13.8 months; HR, 0.56 [95% CI, 0.39 to 0.80]), highlighting the benefit of osimertinib-chemotherapy for patients with high-risk features. OS was consistently superior with the combination of osimertinib and chemotherapy across all subgroup analyses, including patients with and without brain, liver, or bone metastases, different types of EGFR mutations, presence of circulating cell-free DNA at diagnosis, and p53 mutation status.³⁶ The most common AEs with the combination treatment were anemia (46%), diarrhea (43%), and nausea (43%). AEs grade ≥ 3 occurred in 64% (combination) versus 27% (monotherapy), driven by hematologic and GI toxicity. Treatment discontinuation due to AEs occurred in 12% of patients treated in the combination arm, and the median duration of total exposure to pemetrexed was 8.3 months (range, 0.7–58.9), with patients receiving a median of 11 cycles.¹¹

The second study reporting on the benefits of combination therapy over osimertinib monotherapy was MARIPOSA.^{5,35}

In this phase III, international RCT (ClinicalTrials.gov identifier: [NCT04487080](https://clinicaltrials.gov/ct2/show/study/NCT04487080)), 1,074 patients with previously untreated locally advanced or metastatic NSCLC with an EGFR mutation (exon 19 deletion or exon 21 L858R substitution) were randomly assigned in a 2:2:1 ratio to receive amivantamab-lazertinib (n = 429), osimertinib (n = 429), or lazertinib (n = 216). Patients in the combination arm received amivantamab IV (1,050 mg if < 80 kg or 1,400 mg if ≥ 80 kg) once weekly for 4 weeks, then every 2 weeks, plus lazertinib 240 mg orally once daily; osimertinib was given 80 mg once daily and lazertinib 240 mg once daily. The primary end point was PFS in amivantamab-lazertinib group versus osimertinib group. The median PFS was significantly longer in the combination therapy arm compared with osimertinib (23.7 v 16.6 months; HR, 0.70 [95% CI, 0.58 to 0.85]; $P < .001$). Similar to FLAURA2, the ORR was comparable between treatment groups (86% with combination v 85% with osimertinib). The median duration of response (DoR) was longer with the combination compared with monotherapy (25.8 v 16.8 months, respectively). Again, in this study, combination therapy led to prolonged PFS for patients with baseline CNS metastasis (18.3 months v 13.0 months; HR, 0.69 [95% CI, 0.53 to 0.92], respectively). With a median follow-up of 37.8 months, treatment with amivantamab plus lazertinib resulted in a significant OS advantage compared with osimertinib. The HR for death was 0.75 (95% CI, 0.61 to 0.92; $P = .005$). At 3 years, OS rates were 60% for the amivantamab-lazertinib group and 51% for the osimertinib group. Grade ≥ 3 AEs occurred in 75% of patients treated with amivantamab-lazertinib, with AEs leading to treatment discontinuation in 10% (combination arm) versus 3% (osimertinib). Importantly, the toxicity profile of this combination is quite distinct from the FLAURA2 regimen. The most common AEs included paronychia (68%), infusion-related reactions (63%), and rash (62%), with grade 3 AEs driven by skin and/or nail toxicity, embolic events (pulmonary embolism), and infusion reactions each requiring its own prophylactic measures. Other common AEs with this combination include hypoalbuminemia, increased liver enzymes, peripheral edema, and stomatitis.

Clinical interpretation. FLAURA2 and MARIPOSA support the superior efficacy of combination treatment compared to osimertinib alone in the first-line setting for patients with advanced or metastatic NSCLC with EGFR exon 19 deletion or exon 21 L858R substitution. Both combination strategies improve PFS and OS significantly compared to osimertinib monotherapy, although at this time, there is a lack of data directly comparing FLAURA2 to MARIPOSA. The observed benefits of combination therapies must be balanced against a substantial increase in toxicity, infusion time, and treatment frequency. Clinicians should consider these regimens as the preferred treatment option, although osimertinib monotherapy remains an option for selected patients prioritizing tolerability.

TABLE 1. Summary of All Recommendations

Driver Alteration	Recommendation
<i>General Note.</i> The following recommendations (strong or conditional/weak) and terminology (see the Data Supplement) represent reasonable options for patients depending on clinical circumstances and individual patient preferences. Recommended care should be accessible to patients whenever possible. For recommendations with multiple treatment options of the same evidence quality and strength of recommendation, the decision of which agent to offer should be tailored to each patient incorporating both efficacy and toxicity. All biomarkers should be available at the time of decision making.	
Clinical question 1: What are the most effective first-line treatment options for patients' status based on the driver alterations?	
Biomarker testing	1.0. Biomarker testing with a validated tissue and/or blood-based broad multigene panel and a validated tissue IHC assay for PD-L1, HER2, and MET protein expression should be universally accessible for all patients diagnosed with NSCLC. (Evidence quality: High; Strength of recommendation: Strong) <i>Qualifying Statement for Recommendation 1.0:</i> Combination of blood and tissue testing may maximize chance of detecting molecular alterations. Comprehensive NGS testing ideally incorporating RNA-based NGS for detecting genomic alterations is preferred. Tissue testing has the advantage of histologic assessment and should be attempted when feasible, and the false-negative rate of liquid biopsy should be considered. PD-L1 IHC alone should not be used to guide treatment decisions. Treatment decisions based on HER2 overexpression and MET IHC are discussed in the nondriver alteration guideline
EGFR	<i>Exon 19 deletion, Exon 21 L858R substitution</i> 1.1. Clinicians should offer osimertinib with platinum-pemetrexed chemotherapy or amivantamab plus lazertinib. (Evidence quality: Moderate; Strength of recommendation: Strong) 1.2. Clinicians should offer osimertinib to patients not pursuing combination therapy. (Evidence quality: Moderate; Strength of recommendation: Strong) <i>Others</i> 1.3. For other activating <i>EGFR</i> alterations (G719X, L861Q, S768I), clinicians may offer afatinib. (Evidence quality: Moderate; Strength of recommendation: Strong) 1.4. or osimertinib. (Evidence quality: Low; Strength of recommendation: Conditional) <i>Qualifying Statement for Recommendations 1.3 and 1.4:</i> Recommendations 1.3 and 1.4 exclude exon 20 insertion alterations, T790M <i>Exon 20 insertions</i> 1.5. Clinicians should offer chemotherapy plus amivantamab. (Evidence quality: Moderate; Strength of recommendation: Strong) <i>All EGFR Mutations</i> 1.6. For any activating <i>EGFR</i> alteration, regardless of PD-L1 expression levels, single-agent immune checkpoint inhibitors should not be offered as first-line therapy. (Evidence quality: Moderate; Strength of recommendation: Strong)
ALK	1.7. Clinicians should offer alectinib or brigatinib or lorlatinib. (Evidence quality: High; Strength of recommendation: Strong) 1.8. If alectinib, brigatinib, or lorlatinib are not available, clinicians should offer ceritinib or crizotinib. (Evidence quality: High; Strength of recommendation: Strong)
ROS1	1.9. Clinicians may offer crizotinib, entrectinib, repotrectinib, or taletrectinib. (Evidence quality: Moderate; Strength of recommendation: Strong) 1.10. If crizotinib, entrectinib, repotrectinib, or taletrectinib are not available or tolerated, clinicians may offer ceritinib or lorlatinib. (Evidence quality: Low; Strength of recommendation: Conditional)
<i>BRAF</i> ^{V600E}	1.11. Clinicians may offer dabrafenib and trametinib, or encorafenib and binimetinib. (Evidence quality: Moderate; Strength of recommendation: Strong)
<i>MET</i> exon 14 skipping mutation	1.12. Clinicians may offer capmatinib or tepotinib. (Evidence quality: Moderate; Strength of recommendation: Strong)
<i>RET</i> rearrangement	1.13. Clinicians should offer selpercatinib. (Evidence quality: High; Strength of recommendation: Strong) 1.14. If selpercatinib is not available, clinicians may offer pralsetinib. (Evidence quality: Moderate; Strength of recommendation: Strong)
<i>NTRK</i> rearrangement	1.15. Clinicians may offer entrectinib or larotrectinib. (Evidence quality: Moderate; Strength of recommendation: Strong)
General approaches	1.16. For patients with a PS >2, tyrosine kinase inhibitor may be offered based on drug access and toxicity profile and based on clinician evaluations that the benefit outweighs the risk. (Evidence quality: Low; Strength of recommendation: Conditional) 1.17. Patients with advanced lung cancer should be referred to interdisciplinary palliative care teams (consultation) that provide inpatient and outpatient care early in the course of disease, alongside active treatment of their cancer. (Evidence quality: High; Strength of recommendation: Strong)
Clinical question 2: What are the most effective second-line and subsequent treatment options for patients based on the driver alterations?	
NOTE: Due to development of potentially targetable resistance mechanisms and histologic transformation, every effort should be made to assess for presence of histologic transformation by tissue biopsy. Acquired resistance mutations can be checked by tissue and/or blood NGS testing. If patients have received all targeted options, or if no targeted options are available, clinicians may offer standard therapy following the nondriver alteration guideline.	
(continued on following page)	

TABLE 1. Summary of All Recommendations (continued)

Driver Alteration	Recommendation
EGFR	<i>Exon 19 deletion, exon 21 L858R substitution</i>
	2.1. For patients whose cancer develops <i>EGFR</i> T790M resistance alterations in tumor after first- or second-generation <i>EGFR</i> TKIs, clinicians should offer osimertinib. (Evidence quality: High; Strength of recommendation: Strong)
	2.2. For patients who have progressive disease on osimertinib (or other third-generation TKI), clinicians may offer platinum-based chemotherapy with amivantamab. (Evidence quality: Moderate; Strength of recommendation: Strong)
	<i>Qualifying Statement for Recommendation 2.2:</i> For patients whose disease has progressed on <i>EGFR</i> TKIs who are not candidates for amivantamab and chemotherapy, chemotherapy alone ± anti-VEGF-targeting therapy combinations may be offered for those with adenocarcinoma histology and where anti-VEGF therapy is considered safe.
	2.3. For patients who are not candidates for Recommendation 2.2, platinum-based chemotherapy may be offered with or without osimertinib. (Evidence quality: Moderate; Strength of recommendation: Conditional)
	2.4. For patients who have progressive disease on <i>EGFR</i> TKI, anti-PD-(L)1 agents with or without platinum chemotherapy are not recommended. (Evidence quality: High; Strength of recommendation: Strong)
	<i>Qualifying Statement for Recommendation 2.4:</i> Multiple phase III trials have shown no benefit to addition of PD-1 to platinum chemotherapy; however, the role of additional immune combinations, including VEGF, is being explored in trials.
	2.5. For patients whose disease has progressed on osimertinib and platinum chemotherapy, clinicians may offer datopotamab deruxtecan. (Evidence quality: Low; Strength of recommendation: Conditional)
	2.6. For patients who have progressive disease on <i>EGFR</i> TKI with acquired <i>MET</i> amplification, clinicians may offer osimertinib in combination with tepotinib or savolitinib. (Evidence quality: Low; Strength of recommendation: Conditional)
	<i>Exon 20 insertion</i>
ALK	2.7. For patients with an exon 20 insertion alteration who have received prior treatment with platinum chemotherapy, clinicians may offer treatment with amivantamab or sunvozertinib. (Evidence quality: Low; Strength of recommendation: Strong)
	2.8. For patients with an exon 20 insertion alteration who have received prior treatment with platinum chemotherapy and amivantamab, clinicians may offer treatment with sunvozertinib. (Evidence quality: Low; Strength of recommendation: Strong)
ROS1	2.9. For patients who have previously received crizotinib, clinicians should offer alectinib, brigatinib, lorlatinib, or ceritinib. (Evidence quality: Moderate; Strength of recommendation: Strong)
	2.10. For patients who have previously received other <i>ALK</i> inhibitors including alectinib or brigatinib, clinicians may offer lorlatinib. (Evidence quality: Low; Strength of recommendation: Strong)
<i>BRAF</i> ^{V600E}	2.11. For patients who have previously received crizotinib, entrectinib, lorlatinib, or ceritinib, clinicians may offer repotrectinib or taletrectinib. (Evidence quality: Moderate; Strength of recommendation: Strong)
	2.12. For patients who have not received <i>BRAF</i> therapy, clinicians may offer dabrafenib and trametinib, or encorafenib and binimetinib. (Evidence quality: Moderate; Strength of recommendation: Strong)
<i>MET</i> exon 14 skipping mutation	2.13. For <i>BRAF</i> alterations other than <i>BRAF</i> ^{V600E} alterations, clinicians should offer standard therapy following the nondriver alteration guideline. (Evidence quality: Low; Strength of recommendation: Strong)
	2.14. For patients who have not received <i>MET</i> -targeted therapy, clinicians may offer capmatinib or tepotinib. (Evidence quality: Moderate; Strength of recommendation: Strong)
<i>RET</i> rearrangement	2.15. For patients who have not received a <i>RET</i> inhibitor, clinicians should offer selpercatinib or pralsetinib. (Evidence quality: Moderate; Strength of recommendation: Strong)
<i>NTRK</i> rearrangement	2.16. For patients who have not received an <i>NTRK</i> inhibitor, clinicians should offer entrectinib or larotrectinib. (Evidence quality: Moderate; Strength of recommendation: Strong)
<i>HER2</i> exon 20 mutations	2.17. Clinicians may offer treatment with trastuzumab deruxtecan or zongertinib or sevabertinib. (Evidence quality: Low; Strength of recommendation: Strong)
	2.18. For patients who have previously received trastuzumab deruxtecan, clinicians may offer sevabertinib or zongertinib. (Evidence quality: Very Low; Strength of recommendation: Conditional)
<i>KRAS</i> G12C	2.19. Clinicians may offer treatment with sotorasib or adagrasib. (Evidence quality: Moderate; Strength of recommendation: Strong)
	<i>Qualifying Statement for Recommendation 2.17:</i> Note that adagrasib and sotorasib are approved for patients who have received prior chemotherapy and/or anti-PD-(L)1 for patients with advanced <i>KRAS</i> G12C mutation-positive NSCLC. In the first-line setting, these patients should be offered standard first-line treatment with immune checkpoint inhibitor therapy and/or chemotherapy following the nondriver alteration guideline.
<i>NRG1</i> fusion	2.20. Clinicians may offer treatment with zenocutuzumab. (Evidence quality: Low; Strength of recommendation: Strong)

NOTE. The strength of the recommendation is defined as follows: Strong: In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects. In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects. All or almost all informed people would make the recommended choice for or against an intervention. Conditional/weak: In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists. In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists. Most informed people would choose the recommended course of action, but a substantial number would not.

Abbreviations: ALK, anaplastic lymphoma kinase; EGFR, epidermal growth factor receptor; HER2, human epidermal growth factor receptor 2; IHC, immunohistochemistry; MET, mesenchymal epithelial transfection; NGS, next-generation sequencing; NSCLC, non-small cell lung cancer; NRG1, neuregulin 1; NTRK, neurotrophic tyrosine receptor kinase; PS, performance status; RET, rearranged during transfection; ROS1, c-ROS oncogene 1; TKI, tyrosine kinase inhibitor; VEGF, vascular endothelial growth factor.

Recommendation 1.2.

Literature review update. The FLAURA study was a global, double-blind, phase III RCT evaluating first-line osimertinib compared with first-generation *EGFR* tyrosine kinase inhibitors (TKIs; gefitinib or erlotinib) in patients with advanced NSCLC harboring *EGFR* exon 19 deletions or exon 21 L858R substitution. In the final OS analysis (median follow-up: 35.8 months), osimertinib significantly prolonged OS compared to the control arm (38.6 v 31.8 months; HR, 0.80 [95.05% CI, 0.64 to 1.00]; $P = .046$).³⁷ This was accompanied by a substantial improvement in PFS (18.9 v 10.2 months, respectively; HR, 0.46; $P < .001$). Importantly, this benefit was seen across most prespecified subgroups. Among patients without CNS metastasis at baseline, the median PFS with osimertinib was 19.1 months, compared to 10.9 months with first-generation *EGFR*-TKIs (HR, 0.46). In patients with CNS metastasis, median PFS was 15.2 versus 9.6 months, respectively (HR, 0.47). In terms of safety, grade ≥ 3 AEs were reported in 42% of patients receiving osimertinib, compared to 47% in the erlotinib or gefitinib arm. The most common grade ≥ 3 AEs with osimertinib included anemia (3%), decreased appetite (3%), and diarrhea (3%). The discontinuation rate due to AEs was 15% with osimertinib.

Clinical interpretation. Osimertinib remains an appropriate frontline option for select patients, including those with poor PS and those not interested in intensification of therapy. Further, it remains unclear whether a sequential approach could yield comparable long-term outcomes. In these clinical scenarios, monotherapy may offer a more tolerable and effective strategy while preserving future treatment options.

Newly approved targeted agents such as osimertinib have further expanded the therapeutic landscape. However, access to osimertinib varies globally. In settings where osimertinib is not yet available, earlier-generation inhibitors or chemotherapy-based regimens may remain appropriate alternatives, with treatment selection guided by efficacy, toxicity profiles, and regional availability. Beyond osimertinib, other third-generation kinase inhibitors such as almonertinib (aumolertinib),³⁸ lazertinib,^{39,40} and furmonertinib (alflutinib)⁴¹ have shown improved efficacy when compared to first-generation inhibitors and are approved for use in various regions. In regions where access to third-generation kinase inhibitors remains limited, second-generation agents such as afatinib and dacomitinib represent reasonable alternatives to first-generation drugs. However, their use is often constrained by higher toxicity.^{42,43} When newer inhibitors are unavailable, first-generation *EGFR* inhibitors (eg, gefitinib, erlotinib, icotinib) may still provide meaningful clinical benefit, offering superior response rates, tolerability, and PFS compared with chemotherapy.^{17,20} Moreover, combining gefitinib with chemotherapy has been shown in two randomized trials to

improve survival outcomes relative to gefitinib alone, albeit at the cost of increased toxicity.⁴⁴⁻⁴⁶

Others

Recommendations 1.3. and 1.4.

Literature review update. The guideline recommendation remains unchanged, except for the removal of the reference to standard treatments based on the nondriver alteration guidelines as an alternative to targeted therapy. The phase II Japanese UNICORN trial assessed the activity of first-line osimertinib therapy in 40 patients with uncommon *EGFR* mutation NSCLC. The mutations included were G719X (50.0%), S768I (25.0%), and L861Q (20.0%). An ORR of 55% and a median PFS of 9.4 months were noted.⁴⁷ A randomized phase III study in patients with uncommon or compound sensitizing mutations compared afatinib with platinum doublet therapy in the first-line advanced disease setting and showed prolonged PFS (HR, 0.422 [95% CI, 0.256 to 0.694]) and higher ORR with afatinib (61.4% v 47.1%).⁴⁸ This study was reviewed in more detail in the last version of the guideline update.¹ Recent single-arm, real-world, and preclinical studies report that these uncommon mutations are heterogeneous and may exhibit differential sensitivity to specific classes of *EGFR* inhibitors. For example, osimertinib may have greater activity against L861Q, while afatinib may be more potent against G719X.⁴⁹ A new structural classification of atypical mutations has recently emerged, indicating that grouping mutations into structure-function groups rather than exon location may better predict the efficacy of specific classes of *EGFR* TKIs.⁴⁹ Based on this approach, atypical mutations are now divided into P-loop and α C-helix compressing (PACC) and non-PACC (classical-like) mutations. Examples of PACC mutations include G719X, S768I, L747P/S, and V769L, whereas atypical mutations classified as classical-like include L861Q/R and S881F. Although this classification is derived from several large retrospective databases and requires prospective validation, it has already been adopted as a patient selection criterion in several ongoing clinical trials, with prospective outcome data anticipated in the near future.

Clinical interpretation. Atypical *EGFR* mutations make up about 5%-15% of all activating *EGFR* mutations. Structure-based functional classification of these mutations, such as defining a subclass of the so-called PACC mutations, suggests potential differential activity of different generations of TKIs in such tumors.⁴⁹ For atypical but sensitizing mutations including G719X, L861Q, and S768I, afatinib remains the most widely approved drug, with the option of osimertinib based on limited data in this setting with further data needed as to the clinical impact regarding patient selection based on functional classification.⁵⁰ Data on the significant efficacy of combination regimens such as amivantamab-lazertinib for such patients have been reported (ORR of 55% and PFS of 19.5 months), with full data awaited.⁵¹

Recommendation 1.5.

Literature review update. There were no changes to the guideline recommendations. The PAPILLON trial assessed the benefit of adding the EGFR-MET bispecific antibody amivantamab to carboplatin-pemetrexed chemotherapy versus doublet chemotherapy alone in the front-line treatment of patients with advanced NSCLC harboring *EGFR* exon 20 insertion alterations. The doublet chemotherapy-amivantamab combination led to improvements in PFS (HR, 0.40 [95% CI, 0.30 to 0.53]; $P < .001$) compared to chemotherapy alone.⁵² Similarly, response rates were also better with the amivantamab-chemotherapy combination versus doublet chemotherapy alone (73% v 47%; odds ratio, 3.0 [95% CI, 1.8 to 4.8]; $P < .0001$). The safety profile of the combination is comparable to that of each agent alone, and the most common AEs associated with this regimen include neutropenia (59%), paronychia (56%), and rash (54%). Other key amivantamab-related AEs include peripheral edema and infusion reactions.

Clinical interpretation. The OS results of the PAPILLON trial are not yet mature but given the notable and clinically meaningful improvement in both PFS and response rate, amivantamab plus platinum doublet chemotherapy combination is recommended in the first-line setting for patients with advanced NSCLC harboring *EGFR* exon 20 insertion alterations.

Recommendation 1.6.

Literature review update. There were no changes to this recommendation. The lack of benefit of single-agent checkpoint inhibition in managing patients with advanced NSCLC with an *EGFR* mutation was highlighted by a phase II study enrolling patients with PD-L1–positive advanced NSCLC with an *EGFR* mutation. Enrollment was ceased due to lack of efficacy after 11 of 25 planned patients were treated and a 0% response rate was observed in the cohort of patients with confirmed NSCLC with an *EGFR* mutation. In addition to lack of efficacy, an early death due to pneumonitis following switch to erlotinib raised toxicity concerns when patients switch from immunotherapy to *EGFR* TKI therapy.

Clinical interpretation. The panel continues to strongly emphasize the need for appropriate testing before initiating first-line therapy, since patients with alterations in *EGFR* and *ALK* were generally excluded from first-line studies of immunotherapy alone or in combination with chemotherapy,^{53,54} and prospective trials have failed to show significant benefit from immunotherapy alone in patients with *EGFR* alterations and raised concerns as to harm due to excessive toxicity when switching from checkpoint inhibition to *EGFR* TKI therapy.⁵⁵

ALK

Recommendations 1.7 and 1.8.

Literature review update. There were no changes to this recommendation. Clinicians should offer alectinib, brigatinib,

or lorlatinib as first-line treatment. All pivotal first-line trials—ALEX/J-ALEX/ALESIA (alectinib),^{56–58} ALTA-1L (brigatinib),⁵⁹ and CROWN (lorlatinib)⁶⁰—used crizotinib as the comparator. Since the 2023 guidelines,⁶¹ updated analyses have been reported for alectinib and lorlatinib. No additional RCTs were identified since.

The long-term efficacy of alectinib in *ALK*-positive NSCLC has been supported across multiple pivotal studies, including the global ALEX trial,¹² the Japanese J-ALEX trial,⁶² and the Asia-based ALESIA trial⁶³ with extended follow-up. In the final OS update for the global ALEX trial, at a median follow-up of 53.5 (alectinib) and 23.3 (crizotinib) months, median OS was 81.1 months for the alectinib arm versus 57.2 months for the crizotinib arm (HR, 0.78 [95% CI, 0.56 to 1.08]).¹² The 7-year OS rate was 48.6% (95% CI, 40.0% to 57.3%) in the alectinib arm and 38.2% (95% CI, 28.9% to 47.5%) in the crizotinib arm. In the 5 year update from the Asia-based ALESIA trial comparing alectinib to crizotinib, the median PFS with alectinib was 41.6 months (compared to 11.1 months with crizotinib, stratified HR, 0.33 [95% CI, 0.23 to 0.49]).⁶³ OS data remained immature, although the 5-year OS rates were 66.4% for alectinib versus 56.1% for crizotinib. Additionally, the final OS analysis of the Japanese J-ALEX trial was published in 2022 and showed that although therapy with alectinib lead to a significant improvement in the median PFS compared to crizotinib (34.1 months v 10.2 months, HR, 0.37 [95% CI, 0.26 to 0.52]), the final OS analysis after more than 5 years of follow-up did not show superiority of alectinib compared to crizotinib.⁶² In addition to strong long-term efficacy data, the studies also highlight the impressive intracranial disease control with alectinib in patients with *ALK*-driven NSCLC with CNS response rate of 59% ($N = 38$) among 64 patients with baseline CNS metastasis enrolled in the ALEX study.⁶⁴ The safety profile of alectinib was consistent across different studies, with blood bilirubin increase (57%) and blood phosphokinase increase (49%) reported as the most common AEs in the ALESIA study. Constipation (40%) and anemia (28.9%) were reported as most common AE in the global ALEX study, with incidence of grade ≥ 3 AEs ranging from 48% to 58% across studies.

The efficacy of brigatinib in *ALK* inhibitor–naïve, advanced *ALK*-positive NSCLC was established by the phase III ALTA trial.⁶⁵ The study, published in 2021, randomly assigned patients to receive brigatinib or crizotinib. The results showed that after a median follow-up of 40.4 months in the brigatinib arm and 15.2 months in crizotinib arm, brigatinib therapy significantly improved the median PFS compared to crizotinib (30.8 months v 9.2 months; HR, 0.43 [95% CI, 0.31 to 0.58]). The OS data were still immature. In addition to the robust systemic activity, brigatinib shows intracranial activity with a 4-year intracranial PFS rate of 22% in patients with baseline brain metastasis. The most common treatment-related AEs reported with brigatinib were diarrhea (58%) and increased blood creatine phosphokinase (CPK; 50%) with grade ≥ 3 AEs in 78% patients in the brigatinib arm.

In addition to alectinib and brigatinib, lorlatinib has emerged as a highly potent ALK inhibitor with impressive systemic and intracranial efficacy in patients with ALK-driven NSCLC. Lorlatinib was evaluated as the first-line treatment strategy in advanced ALK-positive NSCLC in the phase III CROWN study. A recent 5-year follow-up of CROWN⁶⁶ reported that the median PFS (lorlatinib v crizotinib: not reached [NR] v 9.1 months; HR, 0.19) and the median time to intracranial progression (NR v 16.4 months; HR, 0.06) remained unreached in patients allocated to lorlatinib. Median time to intracranial progression was NR (lorlatinib v crizotinib: NR v 16.4 months; HR, 0.06). Among patients without baseline brain metastases, 96% remained free of new brain metastases at 5 years. The most common AEs with frontline lorlatinib included edema, weight gain, hypercholesterolemia, and hypertriglyceridemia; grade 3–4 AEs occurred in 77% of patients (compared to 57% with crizotinib) and were driven chiefly by hypertriglyceridemia (25%), hypercholesterolemia (21%), weight gain (23%), and hypertension (12%). OS data remain immature.⁶⁶

Clinical interpretation. No randomized head-to-head trials have compared alectinib, brigatinib, and lorlatinib in the first-line setting. While alectinib demonstrates generally favorable tolerability, the data on lorlatinib show a highly durable systemic and CNS control. Given the higher occurrence of grade ≥ 3 AEs and need for proactive monitoring and dose modifications with lorlatinib, the selection among first-line agents should consider CNS risk, reported systemic PFS, cardiac comorbidities, and the distinct AE profiles reported in these trials. Crizotinib or ceritinib should be considered as first-line treatment options in settings where alectinib, brigatinib, or lorlatinib are not available.

ROS1

Recommendations 1.9 and 1.10.

Literature review update. Version 2023.3 of the therapy for stage IV NSCLC with driver alterations living guideline recommended clinicians should offer crizotinib or entrectinib for patients with ROS1 rearrangements. Based on the findings of the TRIDENT-1 trial,⁶⁷ version 2024.1 added repotrectinib as an additional first-line treatment option.²¹ In this guideline update, talrectinib is added as a new option in both first-line and second-line settings.

Talrectinib is an oral, CNS-active, next-generation ROS1 TKI that was studied in locally advanced and metastatic ROS1 mutation-positive NSCLC in TRUST-I (single-institution, Chinese-only)⁶⁸ and TRUST-II (global, multi-institutional) phase II, open-label, single-arm trials.⁶⁹ Talrectinib was administered at 600 mg once daily in both TKI-naïve and TKI-pretreated patients (crizotinib in TRUST-I; crizotinib or entrectinib in TRUST-II). The primary end point was independent review committee–assessed confirmed objective response rate (cORR). Secondary outcomes included intracranial ORR, PFS, DoR, and safety.

In TKI-naïve patients (n = 160), the confirmed ORR was 88.8%, with a median duration of response (DoR) of 44.2 months and a median PFS of 45.6 months. In TKI-pretreated patients (n = 113), the ORR was 55.8%, median DoR 16.6 months, and median PFS 9.7 months. Intracranial ORR was 76.5% in TKI-naïve patients and 65.6% in TKI-pretreated patients with measurable brain metastases. Of the 32 patients with rebiopsy after crizotinib and entrectinib failures, activity was also observed in patients with the G2032R resistance mutation (ORR, 61.5%; 8/13), similar to the activity reported for repotrectinib in TRIDENT-1. See further discussion on TKI-pretreated patients in recommendation 2.9.

The most common AEs were GI-related elevated AST (72%), elevated ALT (68%), diarrhea (64%), nausea (46%), and vomiting (44%). Neurologic AEs occurring in $\geq 10\%$ of patients were dizziness (21%), dysgeusia (15%), and headache (11%); most were grade 1. 1.4% of patients had grade 3 treatment-emergent AEs (TEAEs) with 23.5% dose interruption, 17.5% dose reduction, and only 7% discontinuation.

Clinical interpretation. Given its durable systemic and intracranial efficacy with a favorable safety profile, talrectinib should be considered as one of the first-line treatment options. Neurologic and GI AEs should be considered when choosing treatment in the first-line setting.

If crizotinib, entrectinib, repotrectinib, or talrectinib are not tolerated or unavailable, ceritinib or lorlatinib remain alternatives consistent with previous recommendations.

BRAF^{V600E}

Recommendation 1.11.

Literature review update. This recommendation is unchanged from prior published guidelines. The PHAROS trial included 98 patients (59 treatment-naïve and 39 previously treated) with metastatic NSCLC bearing a BRAF^{V600E} mutation treated with encorafenib and binimetinib.⁷⁰ In treatment-naïve patients, the median PFS was 30.2 months (95% CI, 15.7 to not estimable [NE]). The OS data were recently updated. After a median follow-up for OS of 52.3 months, mOS was 47.6 months (95% CI, 31.3 to NE); 4-year OS probability was 49% (95% CI, 35 to 62).¹³ For all patients, the most frequent grade 3 treatment-related AEs (TRAEs) were AST and ALT increase, diarrhea, nausea, anemia, and asthenia. Any-grade pyrexia occurred in 8% of patients.

Another study by Planchard et al⁷¹ examined the efficacy and safety of dabrafenib and trametinib in 93 patients with metastatic NSCLC harboring a BRAF^{V600E} mutation. This was a phase II, open-label, nonrandomized study. The patients were divided into two cohorts: 36 treatment-naïve patients and 57 previously treated patients. In treatment-naïve patients, the ORR was 64% (95% CI, 46 to 79). The median PFS was 10.9 months (95% CI, 7.0 to 16.6), and the median OS was 24.6 months (95% CI, 12.3 to NE). The most frequent AEs

of any grade were pyrexia, diarrhea, nausea, vomiting, and asthenia. The most common grade 3–4 AEs were pyrexia, AST increase, and ALT increase.

Clinical interpretation. For patients with *BRAF*^{V600E}-positive NSCLC, both dabrafenib + trametinib and encorafenib + binimetinib have demonstrated favorable efficacy and safety as first-line treatment options. Common AEs observed with both combinations include fever, skin disorders, GI disorders, and fatigue, which require careful attention.

In the treatment of cancers with a *BRAF*^{V600E} mutation, there is an ongoing debate about whether to use a TKI or an immunotherapy checkpoint inhibitor as the first-line treatment.

In the lung cancer field, there are not enough data to draw a conclusion yet. The existing information includes a retrospective analysis of 205 patients with *BRAF*^{V600E} mutation-positive NSCLC reported by Wiesweg et al.⁷² The study found no difference in median OS between patients who started with dabrafenib + trametinib (28 months) and those who started with a regimen including an immunotherapy checkpoint inhibitor (27.8 months), with a HR of 1.1.

MET Exon 14 Skipping Mutation

Recommendation 1.12.

Literature review update. There were no changes in this guideline recommendation. Capmatinib is a selective inhibitor of the *MET* receptor. The GEOMETRY mono-1 study, a multicohort (*MET* exon 14 mutation-positive or *MET* amplification according to gene copy number in tumor tissue and based on previous lines of therapy) phase II study, enrolled 364 patients with *MET*-dysregulated advanced NSCLC.⁷³ In patients with *MET* exon 14 mutation-positive disease, ORR was 41% (95% CI, 29 to 53) and 68% (95% CI, 48 to 84) in pretreated and treatment-naïve patients, respectively, with a median DoR of 9.7 months (95% CI, 5.6 to 13.0) and 12.6 months (95% CI, 5.6 to NE), respectively. Responses were rapid and evident at the time of first tumor evaluation in 82% of pretreated patients and 68% of treatment-naïve patients. The median PFS was 5.4 months (95% CI, 4.2 to 7.0) in pretreated patients and 12.4 months (95% CI, 8.2 to NE) in treatment-naïve patients. Seven of 13 evaluable patients with *MET* exon 14 mutation-positive disease and baseline brain metastases demonstrated intracranial responses (4 complete response [CR]), including three patients with prior brain radiotherapy. Furthermore, patients with *MET* amplification (gene copy number ≥ 10) demonstrated an ORR of 29% (95% CI, 19 to 41) and 40% (95% CI, 16 to 68) in pretreated and treatment-naïve patients, respectively. Limited efficacy was observed in pretreated patients with *MET* amplification and gene copy number < 10 . Grade 1–2 peripheral edema (in 51%) and nausea (in 45%) were the most common AEs. Grade 3–4 treatment-related AEs occurred in 37.6% of patients, with the most common being peripheral edema in 8.2% of

patients. Of 13 patients (4%) who died from causes other than advanced NSCLC, one death (pneumonitis) was suspected to be related to capmatinib.

The VISION study was a phase II open-label trial that enrolled 152 patients with *MET* exon 14 mutation-positive treated with tepotinib, with 99 evaluable after ≥ 9 months' follow-up for the primary end point.⁷⁴ The independent-review ORR was 46% (95% CI, 36 to 57) and median DoR 11.1 months (95% CI, 7.2 to NE). Responses were consistent whether *MET* exon 14 mutation-positive was detected by liquid biopsy (ORR, 48%) or tissue biopsy (ORR, 50%) but median DoR was 9.9 months (95% CI, 7.2 to NE) versus 15.7 months (95% CI, 9.7 to NE) in the liquid-biopsy versus tissue-biopsy group, respectively. The median OS (data immature at the time of reporting) was 17.1 months (95% CI, 12.0 to 26.8). Of 11 patients with brain metastases (all nontarget), ORR by independent review was 55% (95% CI, 23 to 83), median DoR was 9.5 months (95% CI, 6.6 to NE), and median PFS was 10.9 months (95% CI, 8.0 to NE). Most common AEs in the study included peripheral edema (63%) and nausea (26%), and most events were grade 1–2. Grade ≥ 3 TRAEs occurred in 28% (including 7% grade 3 peripheral edema) and grade 4 in 2%; discontinuation due to AEs occurred in 11%. TRAEs led to dose reduction in 33% and permanent discontinuations in 11% of patients and were related to peripheral edema (most common), pleural effusion, or dyspnea. One death due to respiratory failure related to interstitial lung disease (ILD) was considered treatment-related.⁷⁴

Crizotinib is associated with a lower response rate (32%) and comparable median PFS. Savolitinib with an ORR of 47.5% and comparable PFS appears promising but detailed information on efficacy and safety signals is unavailable at this time.^{75,76}

Clinical interpretation. Patients who have tumors with a *MET* exon 14 mutation have a poor prognosis with standard therapies, including immunotherapies.⁷⁷ Both capmatinib and tepotinib provide clinically meaningful activity in *MET* exon 14 mutation-positive advanced NSCLC, with ORR approximately 41%–68% for capmatinib depending on line of therapy and approximately 46% with tepotinib across detection modalities; responses are durable and both agents have CNS activity as evidenced by intracranial responses of approximately 55%, supporting use in patients with brain metastases. The safety profile of both agents appears to be manageable, with GI toxicities and peripheral edema being a hallmark toxicity. In practice, choice between agents may be guided by availability, comorbidities, and toxicity management requirements. As in the *BRAF* V600E setting, no randomized clinical trials have compared first-line chemoimmunotherapy with *MET*-targeted therapy. While *MET* TKIs show substantial efficacy in *MET* exon 14 skipping mutations, a subset of patients may achieve durable benefit from immunotherapy. In a retrospective multi-institutional study of 158 patients (80 treated with *MET* TKI and 78 with

immune checkpoint inhibitor $\pm$ chemotherapy), no significant differences were observed in real-world PFS (rwPFS; HR, 0.85; $P = .4$) or OS (HR, 0.97; $P = .9$).

RET Rearrangement

Recommendations 1.13 and 1.14.

Literature review update. This recommendation is unchanged from prior published guidelines. Selpercatinib is a highly selective and potent CNS-active RET kinase inhibitor. In the LIBRETTO-431 phase III RCT, 261 patients with advanced or metastatic *RET* fusion-positive NSCLC were randomly assigned to receive either selpercatinib ($n = 159$) or platinum-based chemotherapy with or without pembrolizumab ($n = 102$). Selpercatinib significantly improved median PFS to 24.8 months compared with 11.2 months for chemotherapy (HR, 0.46 [95% CI, 0.31 to 0.70]; $P < .001$). ORR was 84% (95% CI, 76 to 90) with selpercatinib versus 65% (95% CI, 54 to 75) with chemotherapy. The responses were durable in the selpercatinib group, compared with the control group (24.2 months [95% CI, 17.9 to NE] v 11.5 months [95% CI, 9.7 to 23.3]). In a preplanned subgroup analysis, PFS was longer with selpercatinib than with control across subgroups, including status with respect to intracranial disease at baseline, PD-L1 status, and specific fusion partners (KIF5B, CCDC6, or other). At 21 months of median follow-up time, over 76% of patients were alive in this trial that allowed crossover. OS data remain immature but trend favorably. The most common TRAEs were elevations in AST (61%) and ALT (60%), and hypertension (48%), the majority of which were grade 1-2. Grade ≥ 3 AEs occurred in 28% of patients, most frequently hypertension and transaminase elevations.⁷⁸

Selpercatinib demonstrates notable intracranial efficacy. In the in the LIBRETTO-431 trial, the cause-specific HR for time to CNS progression between groups was 0.28 (95% CI, 0.12 to 0.68), and the 12-month cumulative incidence of CNS progression adjusted for competing non-CNS progression risks and death was 6% versus 20% in the selpercatinib versus control arms, respectively. Of patients who, at baseline, had measurable brain metastases, intracranial response was identified in 82% versus 58% (selpercatinib v control group) with CRs higher in the selpercatinib group versus control group (35% v 17%). In an updated report on the safety and efficacy from the registrational LIBRETTO-001, at a follow-up of 20.2 months, the median DoR was 20.2 months in treatment-naïve patients and 28.6 months in platinum-based chemotherapy pretreated patients. The median treatment duration was 36.1 months.^{79,80}

The ARROW phase I/II multicohort, open-label study enrolled 233 patients with *RET* fusion-positive NSCLC. 92 patients had previous platinum-based chemotherapy and 29 were treatment-naïve. ORR in evaluable pretreated patients was 61% (95% CI, 50 to 71) with 6% CR, and 70% in treatment-naïve patients (95% CI, 50 to 86) with 11% CR.

Shrinkage of intracranial metastases was seen in all nine patients with measurable baseline intracranial metastases. 56% of patients had an intracranial response, and the DoR was NR. Estimates of ongoing intracranial response were 80% at 6 months and projected to be 53% at 12 months. The most common TRAEs ($>20\%$) included elevated transaminases, neutropenia, anemia, asthenia, and constipation; grade 3 or higher TRAEs were neutropenia (18%), hypertension (11%), and anemia (10%) with no treatment-related deaths.⁸¹ Serious TRAEs were reported in 55 (24%) patients and most commonly ($>2\%$) included pneumonia (10 [4%]), pneumonitis (10 [4%]), anemia (five [2%]), and neutropenia (five [2%]). Eighty-nine (38%) patients had dose reductions secondary to TRAEs. Recent information from an ad hoc analysis of the ongoing phase III AcceleRET-Lung study of first-line pralsetinib compared to standard-of-care chemotherapy $\pm$ immunotherapy reported higher rates of severe and fatal infection, including severe opportunistic infections in the pralsetinib arm, prompting a cautionary letter to health authorities by the sponsor.⁸²

Clinical interpretation. Selpercatinib offers significant clinical benefits for patients with *RET* fusion-positive advanced NSCLC by substantially prolonging PFS and achieving high objective response rates compared to chemotherapy in treatment-naïve and platinum-based chemotherapy pretreated populations. It demonstrates intracranial activity with high response rates and impressive reductions in cumulative incidence of CNS metastases when compared to standard of care. While OS data are still maturing, current trends support its benefit with a manageable safety profile.

Pralsetinib also shows promising activity but appears to carry a higher risk for serious AEs (SAEs) including fatal opportunistic infections. Efficacy analysis from the phase III study evaluating it in the first-line setting is pending.

These data emphasize prompt introduction of selective RET inhibitors and strongly support the use of selpercatinib in these patients to improve PFS, induce intracranial responses, and delay development of CNS metastases. The strong responses, including in relation to intracranial metastases, noted with pralsetinib in the ARROW trial support its use if selpercatinib is unavailable or contraindicated for use in this population. Nevertheless, close monitoring would be warranted, given the safety report from the AcceleRET-Lung study at this time.

NTRK Rearrangement

Recommendation 1.15.

Literature review update. There were no changes in this guideline recommendation. Entrectinib is a potent inhibitor of tropomyosin receptor kinase (TRK) A, B, and C with activity against *NTRK* gene fusion-positive solid tumors and CNS activity. An integrated database analysis of three phase I or II clinical trials (ALKA-372-001, STARTRK-1, and

STARTRK-2)⁸³ of adult patients with advanced *NTRK* fusion-positive (*NTRK1*, *NTRK2*, or *NTRK3*) solid tumors, irrespective of prior antineoplastic therapy but without prior TRK inhibitor therapy, and including those with previously treated or asymptomatic brain metastases, was analyzed for the coprimary end points of ORR and DOR. Fifty-four adult patients were included in the integrated efficacy analysis, and 22% and 20% of these patients had baseline CNS disease and baseline brain metastases, respectively. The most represented mutations included *NTRK1* or *NTRK3* fusions and the majority (46%) harbored the *ETV6-NTRK3* fusion. Among the 54 efficacy-evaluable patients, ORR was 57% (7% CR). ORR was 70% in the NSCLC cohort (N = 10). Median DoR for the efficacy-evaluable cohort was 10 months (95% CI, 7.1 to NE) and median OS was estimated to be 21 months (95% CI, 14.9 to NE). 50% of patients with and 60% (10% CR) of patients without baseline CNS metastases experienced response. 50% of patients with baseline brain metastases (N = 54) had intracranial response with median intracranial DoR that was NE (95% CI, 5 to NE) and median intracranial PFS of 14 months (95% CI, 5.1 to NE).

The safety analysis included (1) 68 *NTRK* fusion-positive patients from STARTRK-1, STARTRK-2, and ALKA-372-001 and (2) 355 patients including those from the phase I STARTRK-NG pediatric study that also included patients with *ROS1* and *ALK* gene fusions, who received at least one dose of entrectinib. The most commonly reported serious TRAE was cognitive disorder in the overall safety population. The most common grade 3 or 4 TRAEs in both safety populations were increased weight (10% in the *NTRK* fusion-positive population and 5% in the overall safety-evaluable population) and anemia (12% in the *NTRK* fusion-positive population and 5% in the overall population). Serious TRAEs occurred in 10% of patients in the *NTRK* fusion-positive and 9% in the overall safety population, with the most frequent being nervous system disorders in both populations. Four percent of patients in both safety populations discontinued treatment due to TRAE. None of the six deaths in the *NTRK* fusion-positive safety population or the 20 deaths in the overall safety population were attributed to the treatment.

Larotrectinib was evaluated in a combined analysis of three clinical trials: a phase I adult dose-escalation study (ClinicalTrials.gov identifier: [NCT02122913](#)),⁸⁴ a phase I/II pediatric trial (SCOUT, [NCT02637687](#)),⁸⁵ and a phase II adult and adolescent study (NAVIGATE, [NCT02576431](#)).^{86,87} For the pooled analysis, eligible patients were aged 1 month or older, with locally advanced or metastatic non-CNS primary, TRK fusion-positive solid tumor, who had received standard therapy previously, if available. ORR in 153 patients evaluable for response was 79% (95% CI, 72 to 85), with 16% CR. Responses were 73% and 92% in adult and pediatric patients, respectively. Of the 12 patients with lung cancer, 75% (95% CI, 43 to 95) experienced ORR and the median DoR was NE (95% CI, NE to NE). Although baseline brain imaging was not mandated in asymptomatic patients, 13 patients (8% of 159) were known to have baseline brain metastases, of which six

patients had a lung cancer primary. In a post hoc exploratory analysis of 12 evaluable patients with baseline brain metastases (including multiple tumor types), an objective response was observed in 75% of patients. In three of these patients who had evaluable intracranial disease, there was one CR, one partial response with 46% reduction, and one stable disease with 14% reduction in target tumor size, respectively. Among the 260 safety-evaluable patients treated regardless of TRK fusion status, the most common SAEs included transaminitis (<2%) and nausea (1%), and the most common grade 3 or 4 TRAEs included transaminitis (3%), anemia (2%), and neutropenia (2%). No treatment-related deaths occurred.

Repotrectinib is a next-generation, compact macrocyclic TRK/*ROS1*/*ALK* TKI with activity against other TKIs as well and engineered for enhanced CNS penetration and activity against solvent-front resistance mutations. In the multicenter, single-arm, phase I/II TRIDENT-1 trial,⁸⁸ efficacy in adults with *NTRK* fusion-positive locally advanced or metastatic solid tumors (n = 88) showed confirmed ORR of 58% (95% CI, 41 to 73) in TKI-naïve patients (n = 40) with median DoR NE, per blinded independent central review (BICR) using RECIST v1.1. Baseline CNS assessment was mandated and patients with symptomatic brain metastases were excluded. In the subset of patients with measurable intracranial disease, intracranial responses were observed in 100% of evaluable patients (2/2 in the TKI-naïve cohort) per BICR. The most common adverse reactions (>20%) included dizziness, dysgeusia, peripheral neuropathy, constipation, dyspnea, fatigue, ataxia, cognitive impairment, muscular weakness, and nausea. Grade 3 or more TRAEs occurred in 29% of patients, with grade 3 dizziness in 3% of patients. Recommended dosing is 160 mg once daily for 14 days, then 160 mg twice daily until progression or unacceptable toxicity. On June 13, 2024, the US Food and Drug Administration (FDA) granted accelerated approval for repotrectinib for adult and pediatric (≥12 years) patients with *NTRK* fusion-positive solid tumors that have progressed after treatment or have no satisfactory alternatives.

Clinical interpretation. Pivotal clinical trials evaluating larotrectinib, entrectinib, and repotrectinib demonstrated high levels of systemic and intracranial activity, with rapid onset of response and durable benefit. Toxicity across agents was generally manageable, although safety profiles varied: entrectinib was primarily associated with weight gain and anemia; larotrectinib with transaminitis, anemia, and neutropenia; and repotrectinib with grade ≥3 elevations in gamma-glutamyl transferase, neutropenia, hyponatremia, dyspnea, and CNS-related AEs including dizziness, ataxia, cognitive impairment, muscular weakness, and peripheral neuropathy. The panel did not issue specific recommendations regarding use of repotrectinib in the current version of the guidelines, as efficacy and safety outcomes remain immature and comparatively less well characterized than for larotrectinib and entrectinib. This is predominantly due to the recency of data from the patients with *NTRK* gene

fusion-driven NSCLC enrolled in the phase I/II TRIDENT-1 trial and the absence of a fully published manuscript to facilitate comprehensive assessment. In clinical practice, treatment selection may be influenced by drug availability, patient comorbidities, presence of treated or asymptomatic CNS metastases, and individual toxicity considerations. It should be noted that current evidence is derived largely from single-arm, open-label phase I/II studies with relatively small cohorts reflecting the rarity of NTRK fusion-positive NSCLC, and randomized comparative trials are lacking.

General Approaches

Recommendation 1.16.

Literature review update. There were no changes in this guideline recommendation.

EGFR. Multicenter phase II trials have evaluated the efficacy and safety of gefitinib and osimertinib in patients with NSCLC bearing classical EGFR alterations. A multicenter phase II trial in Japan enrolled 30 patients with an Eastern Cooperative Oncology Group (ECOG) PS of 2 or greater for first-line gefitinib, of which 22 had an ECOG PS of 3 or 4.⁸⁹ The reported ORR was 66% and PFS was 6.5 months. Most patients experienced an improvement in PS and 68% of patients with a pretreatment PS of 3–4 improved to a PS of 0 or 1. One SAE of pneumonitis grade 4 was reported, and a total of 7 grade 3 or higher toxicities were reported, including anemia, pneumonitis, AST or ALT elevation, and anorexia.

First-line osimertinib was evaluated in a multicenter phase II trial and enrolled 30 patients across Japan with a PS of 2–4.⁹⁰ Sixty-three percent of patients experienced a disease response, and the disease control rate (DCR) was 93.3%. Within this cohort, the PFS was 8 months with a 25.4-month OS. Most patients experienced an improvement in PS. Grade 3 or higher AEs were observed in 16 patients. AEs of interest of any grade occurred in six patients with ILD (including 1 grade 5 event), 15 with diarrhea, 11 with rash, and 5 with oral mucositis.

ALK. A multicenter phase II trial evaluated the efficacy of alectinib in 18 patients with advanced ALK mutation-positive NSCLC and a PS of 2–4.⁹¹ Fourteen of these patients were treatment-naïve. An ORR of 72.2% was observed, and PFS in the crizotinib-naïve patients was 16.2 months. PS improved to 0 or 1 in 15 patients. No dose reductions or discontinuation of alectinib was reported for AEs.⁸⁹

Clinical interpretation. Lung cancers bearing a targetable molecular alteration can exhibit rapid and dramatic response to therapy, often improving cancer-associated symptoms. Phase III clinical trials that establish molecularly targeted therapies as standard of care frequently exclude patients with ECOG PS of 2 or greater. Small phase II trials report that patients with EGFR or ALK mutation-positive NSCLC can experience disease response and improvement in PS with targeted therapies. An important

caveat to these data is that PFS is lower than expected from the phase III trials, and patients with poor PS may experience more toxicity.

No trials evaluating the safety and efficacy of agents with TRK-related side effects have been reported in patients with PS of 2 or higher. Clinical judgment and shared decision making around the use of lorlatinib, entrectinib, and repotrectinib in patients with ALK or ROS1 mutation-positive NSCLC will be necessary to determine if the risk-benefit profile of those agents meets the patient's goals of care. Starting one of these therapies at a lower dose or having a low clinical threshold for dose reduction may be warranted. Additionally, agents targeting MET (capmatinib, tepotinib) or BRAF^{V600E} (dabrafenib plus trametinib, encorafenib plus binimetinib) mutations have not been systematically evaluated in patients with PS of 2 or higher. Shared decision making in these situations is recommended, given the side-effect profile of these therapies.

Recommendation 1.17.

Literature review update. A multisite RCT evaluated the effectiveness of palliative care delivered via telehealth compared to in-person care delivery for patients with advanced lung cancer. Twelve hundred and fifty patients were randomly assigned 1:1 to the care delivery methods. Equal levels of patient benefit were noted through both care delivery models, including improved patient quality of life and decreased levels of anxiety and depression.^{92,93}

Clinical interpretation. Palliative care improves multiple quality-of-life outcomes for patients with advanced lung cancer. Telehealth delivery of palliative care does not appear to limit its effectiveness and allows for more patients to access this resource.

SECOND-LINE AND SUBSEQUENT THERAPIES

EGFR

Exon 19 Deletion, Exon 21 L858R Substitution

Recommendation 2.1.

Literature review update. There were no changes in this recommendation. The AURA3 trial established the benefit of osimertinib over platinum-based doublet chemotherapy for patients with advanced EGFR mutation-positive NSCLC upon disease progression on prior EGFR TKI therapy with documented T790M resistance alteration. 419 patients with T790M mutation-positive advanced NSCLC were randomly assigned, in a 2:1 ratio, to osimertinib versus carboplatin or cisplatin and pemetrexed for up to six cycles, with maintenance pemetrexed allowed. The median PFS was significantly longer with osimertinib than platinum-pemetrexed (10.1 v 4.4 months; HR, 0.30 [95%, 0.23 to 0.41; $P < .001$]). Similarly, the ORR was better with osimertinib (71% v 31%). In the osimertinib group, the most commonly reported AEs were diarrhea (41%), rash (34%), dry skin (23%), and

paronychia (22%), while in the platinum–pemetrexed group, it was nausea (49%), decreased appetite (36%), constipation (35%), and anemia (30%). Patients on the osimertinib arm had less grade 3 or higher AEs (23% v 47%).⁹⁴

Clinical interpretation. Although most patients with classical *EGFR* activating mutations receive osimertinib in the first-line setting, some patients receive earlier-generation drugs such as gefitinib and erlotinib. At the time of progression on first- or second-generation *EGFR* TKIs, patients should undergo repeat biomarker testing and, if found to have a T790M mutation, should be offered osimertinib.^{94–97}

Recommendation 2.2.

Literature review update. Version 2023.3 of the therapy for stage IV NSCLC with driver alterations living guideline recommended clinicians should offer platinum-based chemotherapy. Based on the findings of the MARIPOSA 2 trial, version 2024.1 changed it to amivantamab plus carboplatin and pemetrexed. Version 2024.2 stipulated that platinum doublet with or without amivantamab is an appropriate option. This guideline updates this recommendation to platinum doublet with amivantamab. MARIPOSA-2, a phase III study, randomly assigned 657 chemotherapy-naïve patients with *EGFR* exon 19 deletion or L858R mutation-positive advanced NSCLC after disease progression on osimertinib to receive either chemotherapy (carboplatin plus pemetrexed) with amivantamab, chemotherapy plus amivantamab and lazertinib, or chemotherapy alone (randomization schema was 1:2:2).⁴⁰ The dual primary end points are PFS for the chemotherapy–amivantamab arm compared to chemotherapy alone and PFS for the chemotherapy–amivantamab–lazertinib arm compared to chemotherapy alone. To date, the results are mature for only the chemotherapy–amivantamab ($n = 131$) comparison to chemotherapy alone ($n = 263$). The median PFS by BICR was 6.3 months for chemotherapy–amivantamab and 4.2 months for chemotherapy alone; HR, 0.48 (95% CI, 0.36 to 0.64). The investigator-assessed median PFS was 8.2 versus 4.2 months for chemotherapy–amivantamab versus chemo, respectively (HR, 0.41; 95% CI, 0.30 to 0.54). The ORR was 64% for chemotherapy–amivantamab and 36% for chemotherapy alone, and the DoR was 6.9 months for chemotherapy–amivantamab and 5.6 months for chemotherapy alone. Intracranial PFS was 12.5 months for chemotherapy–amivantamab and 8.3 months for chemotherapy alone. The chemotherapy–amivantamab arm had a higher frequency and severity of AEs compared to the chemotherapy alone arm, including the need to interrupt treatment (65 v 33%), reduce the dose of treatment (41 v 15%), or discontinue treatment (18 v 4%). Notable frequent TRAEs in chemotherapy versus chemotherapy–amivantamab group included neutropenia (41 v 55%), nausea (35 v 42%), thrombocytopenia (27 v 44%), and anemia (37 v 35%). Chemotherapy–amivantamab treated patients experienced the following events at a severity of grade 3 or higher: neutropenia (45%), thrombocytopenia (15%), anemia (12%), rash (10%), and infusion reaction (5%). Venous thromboembolism occurred in 10% of chemotherapy–

amivantamab patients compared to 5% in chemotherapy alone.

Clinical interpretation. Carboplatin, pemetrexed, and amivantamab may be offered for patients with disease progression on osimertinib but patients should be aware that side effects may be moderate to severe. Strategies to mitigate rash, infusion reaction, and myelosuppression should be considered.

Recommendation 2.3.

Literature review update. This is a new recommendation for the 2025.2 update. The phase III COMPEL study randomly assigned 98 patients with *EGFR* exon 19 deletion or L858R mutation-positive advanced NSCLC after non-CNS disease progression on first-line osimertinib to receive either chemotherapy (carboplatin plus pemetrexed) with osimertinib or chemotherapy with placebo.⁹⁸ The primary end point was PFS, and all patients underwent CNS imaging at baseline and at all systemic scan time points. The study was closed prior to reaching accrual goals due to slow accrual. The investigator-assessed median PFS was 8.4 months with chemotherapy–osimertinib and 4.5 months with chemotherapy–placebo; HR, 0.43 (95% CI, 0.27 to 0.70). Although not powered for OS, a favorable trend was observed (15.9 v 9.8 months; HR, 0.71 [95% CI, 0.42 to 1.23]), and fewer patients developed new CNS lesions (10% v 27%). The most frequently reported AEs of any grade with osimertinib plus chemotherapy versus placebo plus chemotherapy were anemia (65 v 54%), asthenia (40 v 48%), nausea (35 v 48%), and diarrhea (31 v 10%). Chemotherapy–osimertinib treated patients experienced the following events at a severity of grade 3 or higher: neutropenia (15%), thrombocytopenia (6%), anemia (13%), and asthenia (6%).

Clinical interpretation. The COMPEL study was small but shows a PFS benefit to the continuation of osimertinib with chemotherapy. This regimen may be considered for patients who are eligible for combination therapy and do not have CNS progression while on single-agent osimertinib. It may be appropriate for patients who prefer or require alternatives to more intensive treatment approaches.

Recommendation 2.4.

Literature review update. This became a formal recommendation in the 2024.2 update and is reaffirmed in the current update. Several phase III trials have investigated the benefit of adding of PD-(L)1 inhibitors to chemotherapy for treatment of patients with *EGFR* mutation-positive NSCLC after disease progression following *EGFR* TKIs.^{99–101} The KEYNOTE-789, CheckMate-722, and ORIENT-31 studies investigated combinations of platinum-based doublet chemotherapy with pembrolizumab, nivolumab, and sintilimab, respectively. Among these, ORIENT-31 demonstrated a modest PFS improvement with sintilimab plus chemotherapy compared with chemotherapy alone (HR, 0.72 [95% CI, 0.55 to 0.94]), although the absolute gain was small (median 5.5 v 4.3 months).¹⁰¹ None of these trials showed an OS benefit.

Similarly, studies incorporating antiangiogenic agents with PD-(L)1 blockade plus chemotherapy—such as the phase III IMpower150 and IMpower151 trials—did not show meaningful PFS or OS improvements in the *EGFR* mutation-positive subgroups, although IMpower150 reported a non-significant survival trend (HR, 0.74 [95% CI, 0.38 to 1.46]). More recently, the ORIENT-31 and ATLAS phase III studies combining PD-(L)1 inhibition, vascular endothelial growth factor (VEGF) inhibition, and chemotherapy demonstrated improved PFS (HR, 0.51 [95% CI, 0.39 to 0.67]; and HR, 0.62 [95% CI, 0.45 to 0.86], respectively), but again without OS benefit and with increased toxicity.

Consistent results were observed in the phase III HARMONI trial evaluating the novel PD-1-VEGF bispecific antibody, ivonescimab, plus chemotherapy versus chemotherapy alone in patients with *EGFR* mutation-positive NSCLC after targeted therapy disease progression. The study results were presented at the IASLC's WCLC 2025 and showed that while combination significantly improved PFS (HR, 0.52 [95% CI, 0.41 to 0.66]), it did not meet statistical significance for OS in the primary analysis.

Clinical interpretation. Patients with disease progression on *EGFR* TKIs should not be offered anti-PD-(L)1 therapies with or without platinum chemotherapy. The role of adding VEGF inhibition as combination therapy or bispecific antibody platform has shown modest benefit but with increased risk of toxicity. The addition of chemotherapy to osimertinib post non-CNS progression may be offered to patients who are eligible for combination therapy and do not have CNS progression while on single-agent osimertinib. It may be appropriate for patients who prefer or require alternatives to more intensive treatment approaches.

Recommendation 2.5.

Literature review update. This is a new recommendation for this update. Datopotamab deruxtecan (Dato-DXd) is an antibody-drug conjugate (ADC) that targets TROP2. The treatment was recently FDA-approved under an accelerated mechanism, based on response rate and DoR in a pooled analysis. The TROPION Lung-05 study was a single-arm, open-label study of Dato-DXd in patients with NSCLC and actionable genetic alterations previously treated with chemotherapy and targeted therapy.¹⁰² The TROPION Lung-01 study was a phase III randomized study of Dato-DXd versus docetaxel among previously treated patients with NSCLC.¹⁰³ In a pooled analysis of the *EGFR* mutation-positive patients from these two trials, there were 114 patients with *EGFR* mutations treated with Dato-DXd, 57% of whom had received three or more prior lines of therapy.¹⁶ The ORR was 45%, with median DoR of 6.5 months (95% CI, 4.2 to 8.4). The median PFS was 5.8 months (95% CI, 5.4 to 8.2). The most common AEs were mucositis and stomatitis, affecting 69% of patients, with grade 3 or greater mucositis in 9%. In addition, ocular surface side effects like dry eyes, blurry vision, or keratitis occurred in 32% of patients, with grade 3 in 3%. Pneumonitis was a rare event, only occurring in 4% of patients.

Clinical interpretation. Dato-DXd is an active agent in the arsenal of therapeutics for patients with advanced *EGFR* mutation-positive NSCLC. At the current time, it should be positioned after an *EGFR* TKI and platinum-based chemotherapy have both been given, but optimal sequencing of the increasing number of approved therapeutics for *EGFR* mutation-positive cancers has not been defined.

Another TROP2 ADC sacituzumab tirumotecan was recently studied. In the randomized phase III OptiTROP-Lung04 trial (ClinicalTrials.gov identifier: [NCT05870319](https://clinicaltrials.gov/ct2/show/study?term=NCT05870319)) conducted in China, 376 patients with *EGFR* mutation-positive locally advanced or metastatic nonsquamous NSCLC with progression on prior *EGFR*-TKI therapy were assigned to sacituzumab tirumotecan (sac-TMT; n = 188) or pemetrexed plus platinum-based chemotherapy (n = 188).¹⁴ At a median follow-up of 18.9 months, median PFS was 8.3 months (95% CI, 6.7 to 9.9) with sac-TMT versus 4.3 months (95% CI, 4.2 to 5.5) with chemotherapy (HR, 0.49). Interim OS analysis showed a significant benefit favoring sac-TMT (median NR v 17.4 months; HR 0.60, 95% CI, 0.44 to 0.82). The most common AEs associated with sac-TMT included anemia (84.6%), white cell decrease (83.5%), and alopecia (83.5%). Grade ≥3 TRAEs occurred in 58.0% with sac-TMT and 53.8% with chemotherapy, and the most common severe AE was neutropenia (39.9% v 33.0%). No treatment-related discontinuations occurred in the sac-TMT group. Stomatitis (64.4%, grade ≥3: 4.8%) and ocular toxicity (9.6%, all grade 1-2) were more common with sac-TMT.¹⁴ While this monotherapy is an effective treatment option for patients with *EGFR*-TKI-resistant, *EGFR* mutation-positive NSCLC, consideration in interpreting and generalizability of these results should be taken into account, as this study was performed in a non-US population.

Recommendation 2.6.

Literature review update. This is a new recommendation for this update. MET amplification has emerged as a frequent and potentially actionable acquired resistance mechanism on *EGFR* TKI therapy with higher frequency among patients on third-generation TKI therapy. Multiple studies have been assessing the clinical utility of MET targeting in such patients with a series of these studies reporting data recently. The phase Ib TATTON trial¹⁰⁴ suggested promising response rates of 33%-67% for the combination of savolitinib (initially 600 mg once daily then later 300 mg twice daily) and osimertinib 80 mg once daily in cohorts of patients with advanced *EGFR* mutation-positive NSCLC with MET amplification or overexpression following prior *EGFR* TKI therapy. Better antitumor activity was noted among patients with MET gene copy number ≥10. One module of the phase II ORCHARD trial¹⁰⁵ assessed the efficacy of the combination of osimertinib (80 mg once daily) and savolitinib (300 or 600 mg once daily) in 32 patients who progressed on first-line osimertinib with a documented MET gene amplification (≥4 copies of MET over tumor ploidy) with a primary end point of investigator-assessed response rate. An ORR of 47% (80% CI, 35 to 60) was noted along with DoR of 14.5 months

(95% CI, 5.6 to 18.7) and median PFS of 7.6 months (95% CI, 3.2 to 15.9). The most common TEAEs were nausea (44%), fatigue (31%), diarrhea (25%), and arthralgia (25%). Grade ≥ 3 TEAEs included pneumonia (9%), ALT increase (6%), and decreased neutrophil count (6%). The phase II SAVANNAH trial⁸ then enrolled patients after disease progression on osimertinib and results have been now reported on the primary efficacy population of 80 patients—these patients had tumors with MET IHC 3+ or $\geq 90\%$ or fluorescence in situ hybridization (FISH) >10 —who then went on to receive savolitinib orally 300 mg twice daily and osimertinib 80 mg once daily with an ORR of 56% (95% CI, 45% to 67%) and PFS of 7.4 months (95% CI, 5.5 to 7.6). In a randomly assigned subset of 73 patients assessing osimertinib + savolitinib versus savolitinib alone (2:1 randomization), ORR was 58% with a PFS of 8.3 months (95% CI, 5.8 to 15.1) for the combination regimen, while savolitinib alone had an ORR of only 16% and PFS of 3.7 months (95% CI, 1.4 to 5.8). In addition, in patients with evaluable CNS disease, the combination regimen achieved a response rate of 43% versus 25% with single-agent savolitinib. The most frequently reported TEAEs with the combination regimen were peripheral edema (46%), nausea (41%), and diarrhea (23%), while the most common grade ≥ 3 TEAEs were peripheral edema (4%) and pulmonary embolism (4%). Lastly, the phase II single-arm INSIGHT2 trial¹⁰⁶ assessed the utility of osimertinib + tepotinib combination therapy in patients with advanced EGFR mutation-positive NSCLC who had disease progression following first-line osimertinib therapy. Biomarker requirement was MET gene copy number ≥ 5 or MET/CEP7 ratio of ≥ 2 by FISH or MET plasma gene copy number by circulating tumor DNA NGS of ≥ 2.3 . Patients received oral tepotinib at 500 mg and osimertinib 80 mg once daily. The study enrolled 128 patients with a primary end point of ORR in the primary activity analysis population including 98 patients with MET amplification confirmed by central FISH testing who had at least 9 months of follow-up. The cORR in this cohort was 50% (95% CI, 39.7 to 60.3) with a DoR of 8.5 months (95% CI, 6.1 to NE) and a median PFS of 5.6 (95% CI, 4.2 to 8.1) months. Common AEs included diarrhea (49%), edema (41%), paronychia (23%), nausea (21%), and decreased appetite (20%), while the most common treatment-related grade 3 or worse AEs were peripheral edema (5%), decreased appetite (4%), prolonged QTc (4%), and pneumonitis (3%).

Clinical interpretation. MET amplification is a frequent acquired resistance mechanism occurring in 10%–15% of patients with EGFR mutation-positive NSCLC progressing on frontline, third-generation EGFR TKI therapy. Tissue-based FISH, NGS, and IHC, as well as ctDNA-based NGS testing, have been utilized for detection with variable cut-offs. The above-listed clinical trials have highlighted the significant clinical activity of EGFR/MET TKI combination therapy. Further corroboration of such an approach came recently from the randomized phase III SACHI trial in China of osimertinib 80 mg once daily + savolitinib (600 mg once daily for patients >55 kg and 400 mg for patients <55 kg)

versus platinum-pemetrexed doublet chemotherapy (including pemetrexed maintenance) in patients with EGFR mutation-positive NSCLC who had disease progression on prior EGFR TKI therapy (stratified for first- or second-generation v third-generation TKI) with a primary end point of PFS by investigator.¹⁰⁷ The study randomized 211 patients (65% post first- or second-generation; 35% post third-generation TKI) with different biomarker entry criteria used for the two key patient subsets—MET copy number ≥ 5 or MET/centromere enumeration probe (CEP7) ≥ 2 was needed post first- or second-generation TKI, while MET gene copy number ≥ 10 was needed post third-generation TKI. The study results are positive for intention to treat PFS, with a PFS HR of 0.34, median PFS of 8.2 months (95% CI, 6.9 to 11.2) versus 4.5 (3.0 to 4.5), and ORR of 58% versus 34%. Broad benefits of the experimental arm included key subsets of patients including patients with CNS disease and prior third-generation TKI. Grade ≥ 3 AEs were 45% in the osimertinib + savolitinib and 48% in the chemotherapy arm. In light of published data and the recently presented SACHI trial, a combination of osimertinib and an available MET TKI may be a prudent option for patients with documented MET amplification upon progression on osimertinib. Ongoing additional trials, such as the phase III SAFFRON trial (ClinicalTrials.gov identifier: [NCT05261399](https://clinicaltrials.gov/ct2/show/study?term=NCT05261399)), will add further information as to the clinical impact of the targeted combination approach. Significant AEs of such combinations include both EGFR- and MET-related side effects but can be typically mitigated by dose interruptions and dose reductions. Varying biomarker criteria for these studies must be highlighted with more robust benefits in general noted when using more stringent criteria of MET amplified status. Overall, these data underscore the continued need to consider repeat biomarker testing for actionable acquired resistance mechanisms, including MET amplification upon disease progression on targeted therapy.

Exon 20 Insertion

Recommendation 2.7.

Literature review update. Amivantamab after disease progression on platinum-based therapy was recommended in version 2023.3 update. This update adds sunvozertinib as another treatment option. After disease progression on platinum-based therapy without amivantamab, patients with an EGFR exon 20 mutation are appropriate for consideration of targeted therapy. The CHRYSALIS phase I study examined the bispecific EGFR–MET monoclonal antibody amivantamab among 81 patients with previously treated EGFR exon 20 insertion mutation-positive NSCLC with a median of two prior lines of therapy, and found an ORR of 40% and a median PFS of 8.3 months, making it an appropriate option for patients who had not previously received amivantamab.¹⁰²

In addition, sunvozertinib was recently FDA approved at 200 mg once daily dose for the management of patients with *EGFR* exon 20 insertion mutation-positive NSCLC.¹⁰⁸ It is an oral, irreversible, and selective TKI with significant efficacy as demonstrated, for example, in the WU-KONG6 study focusing on a Chinese patient population with platinum pretreated *EGFR* exon 20 insertion mutation-positive NSCLC. In this trial enrolling 104 patients, an ORR of 61% was observed with sunvozertinib 300 mg once daily along with an acceptable safety profile, with CPK elevation, diarrhea, and anemia being the most common grade 3 AEs, while ILD as a SAE was reported in 5% of the patients.⁴ Furthermore, the WU-KONG1B trial randomly assigned previously treated patients with *EGFR* exon 20 insertion mutation-positive NSCLC to 200 mg or 300 mg of sunvozertinib once daily with additional patients treated at the 300 mg per day dose following a predefined interim analysis. 45.9% (200 mg), 47.2% (300 mg randomized), and 45.8% (300 mg all) ORRs were noted demonstrating efficacy at both dose levels. DoR was 11.1 months (200 mg), 13.8 months (300 mg randomized), and 9.8 months (300 mg all). The most common TRAEs at the 200 versus 300 mg dose level included diarrhea (68 v 83%), blood CPK increase (35 v 52%), and rash (41 v 48%). Most common treatment-related grade 3 or higher AEs at 200 versus 300 mg dose levels were diarrhea (2.2% v 18%), blood CPK increase (6.6% v 12.6%), and anemia (4.4% v 6.3%).

Clinical interpretation. Both amivantamab and sunvozertinib are active agents with disparate AE profiles for patients with *EGFR* exon 20 insertion-positive NSCLC upon disease progression on prior platinum-based chemotherapy without amivantamab. Proper sequencing of the available and emerging treatment options for *EGFR* exon 20 insertion-positive NSCLC is to be defined.

Recommendation 2.8.

Literature review update. After disease progression on platinum-based therapy with amivantamab, patients with an *EGFR* exon 20 mutation are appropriate for consideration of further targeted therapy with sunvozertinib as discussed in the WU-KONG1B trial described in Recommendation 2.6.⁴

Clinical interpretation. Sunvozertinib is an active agent with an acceptable safety profile for patients with *EGFR* exon 20 insertion-positive NSCLC upon disease progression on prior platinum-based chemotherapy with amivantamab. Proper sequencing of the available and emerging treatment options for *EGFR* exon 20 insertion-positive NSCLC is to be defined.

ALK

Recommendation 2.9.

Literature review update. There were no changes in this recommendation. Patients with advanced *ALK* mutation-positive NSCLC who experience disease progression on crizotinib should be treated with a second- or third-generation *ALK* TKI with established efficacy in the post-

crizotinib setting. Among the second-generation *ALK* inhibitors, ceritinib was evaluated in the phase III ASCEND-5 trial, which enrolled patients with disease progression on crizotinib and platinum doublet chemotherapy. In the study, ceritinib demonstrated a median PFS of 5.4 months compared to 1.6 months with chemotherapy (PFS inhibitors, HR, 0.49 [95% CI, 0.36 to 0.67]) and intracranial ORR of 36%,¹⁰⁹ demonstrating superiority over single-agent chemotherapy. In a similar setting, alectinib also demonstrated superior outcomes to single-agent chemotherapy with a median PFS of 9.6 months (PFS HR, 0.32; 95% CI, 0.17 to 0.59) and intracranial ORR of 54%.¹¹⁰ Alectinib provides strong systemic and CNS efficacy, and a favorable safety profile, especially in patients with comorbidities or poor PS. Multiple systematic reviews and network meta-analyses rank alectinib as having the lowest incidence of grade 3-4 AEs, with rates of 16%-37%, and the best overall tolerability in the post-crizotinib setting.^{111,112} A phase II study of 18 patients with poor PS (2, 3 and 4) showed a median PFS of 16 months with low frequency of AEs, especially grade 3.¹¹³ Alectinib is effective against several crizotinib-resistant *ALK* mutations, notably L1196M, G1269A, and I1171T, but is less effective against G1202R, a common mechanism of resistance after crizotinib.¹¹⁴

Likewise, brigatinib demonstrated benefit in the post-crizotinib setting, with a median PFS of 12.9 months and an intracranial ORR of 67% in one phase II trial,¹¹⁵ and a median PFS of 16.7 months and an intracranial ORR of 67% with brigatinib 180 mg once daily in the phase II ALTA trial.¹¹⁶ Brigatinib provides durable efficacy but the most common any-grade AEs were nausea (33%), fatigue (29%), headache (28%), dyspnea (27%), and vomiting (24%); grade 3-4 TRAEs in ALTA included elevated CPK (11% in the 180 mg arm), hypertension (4%), increased lipase (4%), and pneumonitis (4%), with early-onset ILD or pneumonitis notably within the first 7-8 days. Close monitoring remains essential, especially for patients with pulmonary or cardiovascular comorbidities. In a Bayesian network meta-analysis,¹¹⁷ brigatinib is associated with a higher incidence of grade 3-4 AEs compared to alectinib, with rates of 63%-73%, but its overall safety profile is still considered acceptable for most patients. Importantly, brigatinib maintains substantial activity against a wide array of *ALK* resistance mutations, including L1196M, G1269A, I1171T/N, and, importantly, G1202R, although its activity against compound mutations is limited.

Although ceritinib is the only agent evaluated in a phase III setting, response rates noted with alectinib and brigatinib (intracranial ORR approximately 50%) in the phase II studies were numerically higher than that observed with single-agent ceritinib (approximately 36%). Although all three agents exhibit intracranial efficacy, alectinib exhibits the most favorable toxicity profile. Ceritinib remains an option where other agents are unavailable or contraindicated but generally carries lower efficacy rates and significant hepatic and GI side effects. The most

common AEs with ceritinib included GI events (diarrhea, nausea, vomiting) and transaminase elevations; grade 3–4 AEs included increased ALT (21%), increased γ -glutamyltransferase (21%), and increased AST (14%). Evidence supporting lorlatinib in patients who have experienced disease progression on both first- (crizotinib) and second-generation (alectinib or brigatinib) ALK TKIs is supported by a single-arm, multicohort phase II trial where patients with advanced ALK mutation-positive NSCLC were divided across six expansion cohorts (EXP1–6) stratified by prior therapy. In previously treated ALK mutation-positive patients (EXP2–5, $n = 198$), systemic ORR was 47.0% (95% CI, 39.9 to 54.2) and intracranial ORR among those with measurable CNS disease ($n = 81$) was 63.0% (95% CI, 51.5 to 73.4). Subgroup analysis demonstrated an ORR of 69.5% (95% CI, 56.1 to 80.8) after crizotinib only ($n = 59$), 32.1% (95% CI, 15.9 to 52.4) after one previous noncrizotinib ALK TKI ($n = 28$), and 38.7% (95% CI, 29.6 to 48.5) after ≥ 2 prior ALK inhibitors ($n = 111$). Corresponding intracranial ORRs were 87.0%, 55.6%, and 53.1%, respectively.¹¹⁸ Lorlatinib offers the broadest resistance mutation coverage with activity against G1202R as well as compound mutations, where it should be considered the agent of choice. It also has extremely high CNS activity and should be considered in patients with marked CNS progression. However, its challenging toxicity profile with hypercholesterolemia and neurocognitive disorders necessitates careful patient selection and monitoring.

The Expert Panel recommends that, at the time of disease progression, repeat genomic testing through plasma or tissue-based NGS should be performed to guide therapy through identification of potential mechanisms of on-target resistance that may confer sensitivity to a particular ALK TKI (eg, G1202R, L1196Q, L1171X).^{119–121}

Clinical interpretation. In the post-crizotinib setting, alectinib, brigatinib, lorlatinib, or ceritinib may all be considered. There are no head-to-head RCTs between second-generation TKIs post-crizotinib, and sequencing is highly individualized. Drug selection should consider several factors including efficacy (systemic and CNS), resistance mutation profile, safety and tolerability, comorbidities, and patient-centered factors.

Recommendation 2.10.

Literature review update. This guideline recommendation is unchanged. As described in Recommendation 2.8, lorlatinib shows meaningful activity after second-generation ALK TKIs. In a single-arm, multicohort phase II study, patients previously exposed to ≥ 1 noncrizotinib ALK inhibitor had systemic ORRs of 32.1% after one prior second-generation TKI ($n = 28$) and 38.7% after ≥ 2 prior ALK TKIs ($n = 111$); intracranial ORRs among those with measurable CNS disease were 55.6% and 53.1%, respectively.¹¹⁸ These data support lorlatinib as an appropriate next-line option after alectinib or brigatinib.

After disease progression on lorlatinib, patients should be transitioned to platinum-pemetrexed-based chemotherapy, with consideration of local therapy for oligoprogression. Clinical trials are also ongoing with next-generation (fourth generation) ALK inhibitors with activity against compound ALK mutations that commonly emerge after lorlatinib (eg, NVL-655, TPX-0131).^{122,123}

Clinical interpretation. Given its significant intracranial efficacy and systemic activity even after exposure to one or multiple ALK inhibitors, lorlatinib represents an appropriate option for patients previously treated with a second ALK TKIs such as alectinib or brigatinib. After disease progression on lorlatinib, patients should be transitioned to platinum-pemetrexed-based chemotherapy, with consideration of local therapy for oligoprogression.

ROS1

Recommendation 2.11.

Literature review update. Version 2023.3 of the therapy for stage IV NSCLC with driver alterations living guideline recommended clinicians should offer lorlatinib. In version 2024.1, repotrectinib was added. In this guideline update, taletrectinib is added as a new option in second-line settings. Two next-generation ROS1 inhibitors—repotrectinib and taletrectinib—have demonstrated promising efficacy and intracranial activity in patients with ROS1 mutation-positive NSCLC with prior exposure to ROS1 TKIs.^{10,67,69} The post-TKI cohort in the earlier-described TRUST-I and TRUST-II study (refer to Recommendation 1.9) comprised patients previously treated with one approved ROS1 TKI (crizotinib or entrectinib), with or without up to one prior line of systemic chemotherapy. Among TKI-pretreated patients ($n = 113$), taletrectinib achieved a cORR of 55.8% with a median DoR of 16.6 months and a median PFS of 9.7 months. Intracranial ORR was 65.6% in those with measurable baseline brain metastases. Responses were seen in G2032R mutation-positive disease with a cORR of 61.5% (8/13). Prior TKI exposure was $>90\%$ (103/113) crizotinib and approximately 9% (10/113) entrectinib. By prior TKI exposure, cORR was 53.4% after crizotinib ($n = 103$) and 80.0% after entrectinib ($n = 10$). Neurologic AEs were infrequent and largely grade 1. The most common AEs were GI-related elevated AST (72%), elevated ALT (68%), diarrhea (64%), nausea (46%), and vomiting (44%). Neurologic side effects were infrequent (dizziness, 21%; dysgeusia, 15%).¹⁰

The post-TKI cohort in the TRIDENT-1 trial of repotrectinib included patients treated with one ROS TKI and chemotherapy ($n = 56$). The confirmed ORR was 38% (21/56), with a median DoR 14.8 months and a median PFS 9.0 months; activity was retained in solvent-front mutation-positive disease (G2032R mutation: 10/17 responses, 59%). Intracranial activity in the pretreated setting was 41.7% (5/12 responses) among patients with measurable baseline brain metastases, and 33% (8/24) when both measurable and nonmeasurable lesions were considered. Prior TKI exposure

in this cohort was predominantly crizotinib (82%, 46/56), entrectinib (16%, 9/16), and ceritinib (2%, 1/56). Neurologic side effects included dizziness in 58% of patients, dysgeusia in 50%, and paresthesia in 30%.¹²⁴ Rebiopsy at the time of disease progression is recommended standard clinical practice in *ROS1* mutation-positive NSCLC. Both tissue and liquid biopsies can characterize resistance mechanisms, with the detection of on-target kinase domain mutations, bypass pathway alterations, and other genomic events that inform subsequent therapy selection. In *ROS1* mutation-positive NSCLC, rebiopsy frequently uncovers on-target resistance mutations (eg, G2032R, D2033N, L2086F, S1986F/Y) and off-target alterations such as *KRAS* alteration or *MET* amplification. Liquid biopsy has been shown to have adequate specificity for both fusions and point mutations in this setting, especially if tissue is scarce or unavailable due to procedural risks and is less likely to be affected by tumor heterogeneity.¹²⁵ The solvent-front mutation G2032R is the most frequently identified on-target resistance mutation in *ROS1* in NSCLC after disease progression on first-generation TKIs such as crizotinib and entrectinib. Both repotrectinib and taletrectinib retain significant systemic activity against G2032R. The L2086F mutation, located in the central β -sheet 6 region, confers resistance to all currently approved next-generation *ROS1* TKIs, including repotrectinib and taletrectinib.¹²⁶ Preclinical data and case reports indicate promising activity for cabozantinib in this setting.¹²⁶ Off-target bypass alterations (eg, *KRAS* or *MET* amplifications) are unlikely to respond and may require alternative systemic strategies or combination approaches. Thus, molecular reassessment is integral to optimizing sequencing decisions.

Clinical interpretation. After prior *ROS1* TKI, either repotrectinib or taletrectinib may be offered. Both agents show activity in G2032R mutation-positive disease and intracranial metastases, with broadly similar performance. Molecular testing should be considered at the time of progression to optimize sequencing.

Selection among repotrectinib and taletrectinib should consider prior TKI exposure, expected adverse event patterns (CNS events more frequent with repotrectinib; GI events more frequent with taletrectinib), CNS activity (intracranial ORR of 65% for taletrectinib and 38% for repotrectinib) and potential drug-drug interactions, and availability.

BRAF^{V600E}

Recommendations 2.12 and 2.13.

Literature review update. There were no changes to this recommendation. In patients with advanced NSCLC harboring a *BRAF*^{V600E} mutation who have experienced disease progression on nontargeted therapy, targeted combination therapy with a *BRAF* and MEK inhibitor has demonstrated meaningful clinical benefit. The PHAROS trial investigated encorafenib and binimetinib in 98 patients with metastatic NSCLC who had a *BRAF*^{V600E} mutation. The study enrolled two

groups: 59 patients who had not been previously treated and 39 patients who had received prior treatment.⁴⁶ In the previously treated cohort, the median PFS was 9.3 months (95% CI, 6.2 to 24.8). The ORR was 46% (95% CI, 30 to 63). Median follow up for OS was 48.2 months, resulting in mOS of 22.7 months (95% CI, 14.1 to 32.6); 4-year OS probability was 31% (95% CI, 16 to 47).¹³

Planchard et al investigated the effectiveness and safety of dabrafenib and trametinib in 93 patients with metastatic NSCLC who had a *BRAF*^{V600E} mutation.⁴⁷ This was a phase II, open-label, nonrandomized trial, and the patients were split into two groups: 36 who had not received prior treatment and 57 who had been previously treated. In the previously treated cohort, the ORR was 63.2% (95% CI, 49.3 to 75.6). The median PFS was 9.7 months (95% CI, 6.9 to 19.6), and OS data were immature at the time of the presentation.

For patients with metastatic NSCLC harboring *BRAF* alterations other than *BRAF*^{V600E}, there are currently no data to support the use of *BRAF* targeted therapies described above, and these patients should be offered standard therapy following the nondriver alteration guidelines.

Clinical interpretation. In patients with *BRAF*^{V600E} mutation-positive NSCLC who have not previously received *BRAF*-targeted therapy, either dabrafenib + trametinib or encorafenib + binimetinib, both recommended as first-line options, have demonstrated favorable efficacy and safety. Common AEs observed with both combinations include fever, skin disorders, GI disorders, and fatigue, which require careful attention. For patients with *BRAF* alterations other than *BRAF*^{V600E}, standard therapy following the nondriver alteration guidelines should be offered.

MET Exon 14 Skipping Mutation

Recommendation 2.14.

Literature review update. There were no changes to this recommendation. Both capmatinib and tepotinib have demonstrated clinically meaningful and durable activity in *MET* exon 14 mutation-positive NSCLC and should be offered in patients with *MET* exon 14 mutation-positive NSCLC who have not received targeted therapy as their initial treatment. In the GEOMETRY mono-1 study, a multicohort (*MET* exon 14 mutation-positive or *MET* amplification according to gene copy number in tumor tissue and based on previous lines of therapy) phase II study enrolled 373 patients with *MET*-dysregulated advanced NSCLC (160 patients with *MET* exon 14 mutation).^{73,127} The final outcomes of the study demonstrated an ORR of 44% (95% CI, 34.1 to 54.3) among the 100 patients in the pretreated cohort (cohort 4 and 6) with one patient achieving a CR. As per the BIRC, most responses occurred within 2 months of initiating therapy irrespective of treatment lines. The median DoR was 9.7 months (95% CI, 5.6 to 13.0) with a median PFS of 5.5 months (95% CI, 4.2 to 8.1) and a

median OS of 16.8 months (95% CI, 11.6 to 23.8) in pretreated populations. For patients in the pretreated cohort, ORR of 52% (95% CI, 33.1 to 69.8) versus 41% (95% CI, 28.9 to 53.1), DCR of 90% (95% CI, 74.2 to 98.0) versus 78% (95% CI, 66.7 to 87.3), median PFS of 6.9 months (95% CI, 4.2 to 13.3) versus 5.4 months (95% CI, 4.2 to 7.0), and median OS of 26.0 months (95% CI, 13.5 to 43.4) versus 13.6 months (95% CI, 8.6 to 22.2) were superior in the patients who received capmatinib in the second-line setting (cohort 4) when compared to those who received it in the second- or third-line setting (cohort 6). 57% of 28 evaluable patients had intracranial complete or partial responses with 8 of 9 CR in the second-line or third-line setting. 15 of these 28 patients had no previous intracranial radiotherapy, and 67% demonstrated intracranial CRs or partial responses, with all 5 CR occurring in the second-line or third-line setting. The most common TRAEs were peripheral edema (47%), nausea (35%), increased creatinine (21%), and vomiting (20%); 14% patients had SAEs suspected to be treatment-related, of which 10% were grade 3-4. Treatment-related SAEs included vomiting (2%), nausea (1%), and peripheral edema (1%). Four on treatment deaths were assessed as treatment-related and included cardiac arrest, hepatitis, organizing pneumonia, and pneumonitis.¹²⁷

In the VISION study, 313 patients diagnosed with *MET* exon 14 mutation-positive NSCLC were enrolled between the primary (A) and confirmatory (C) cohorts. With a median follow-up of 32.6 months between the two cohorts, treatment with tepotinib demonstrated an ORR of 51.4%, a median DoR of 18.0 months, a median PFS of 11.2 months, and a median OS of 19.6 months across treatment lines. Based on prespecified analyses, previously treated patients ($n = 149$) had an ORR of 45.0%, a median DoR of 12.6 months, a median PFS of 11 months, and a median OS of 19.3 months, with improved outcomes when used in the first-line setting. In 15 patients with target brain metastases evaluable by Response Assessment in Neuro-Oncology Brain Metastases criteria, intracranial ORR was 66.7% (12 patients had previous radiotherapy for brain metastases). Peripheral edema was reported as a TRAE in 67.1% of patients, with 11.2% experiencing grade ≥ 3 edema. The time to onset of this AE was approximately 8 weeks, and although slow to resolve, it was identified to be manageable. GI AEs such as nausea (26.7%), vomiting (12.9%), and diarrhea (26.3%) were generally mild to moderate, occurred early, and resolved within the first few weeks of treatment. The earlier publication detailed that TRAE resulted in dose reductions in 33% and permanent discontinuation in 11% of patients, with one death due to respiratory failure related to ILD considered treatment-related.⁷⁴

Recently published results from the *MET* exon 14-mutant cohort of the phase I CHRYSALIS study included both treatment-naïve and previously treated patients.¹²⁸ The study evaluated amivantamab in 97 participants (16 treatment-naïve, 28 previously treated but *MET* TKI-

naïve, and 53 with prior *MET* TKI exposure). Among previously treated patients, those without prior *MET* TKI therapy achieved an ORR of 46%, whereas patients with prior *MET* therapy exposure had a lower ORR of 19%. No new safety signals were observed, and treatment-related serious or grade ≥ 3 AEs occurred in 35% and 48% of patients, respectively. The small sample size limits definitive conclusions regarding the magnitude of clinical benefit. While treatment options remain limited for patients whose disease progresses after *MET* TKI therapy, the modest efficacy of amivantamab must be balanced against its toxicity risk.

Clinical interpretation. Patients who have tumors with a *MET* exon 14 mutation-positive have a poor prognosis with standard therapies, including immunotherapies.⁷⁷ Both capmatinib and tepotinib provide clinically meaningful activity in the second or subsequent line therapy of patients with *MET* exon 14 mutation-positive advanced NSCLC who did not receive targeted therapy as their initial line of therapy. In this population of patients, capmatinib and tepotinib achieved comparable ORR of 44%-45% with durable responses and robust intracranial activity. The safety profile of capmatinib and tepotinib appear manageable, with peripheral edema being a hallmark toxicity. In practice, patients with *MET* exon 14 mutation-positive NSCLC with disease progression on first-line therapies can be offered capmatinib or tepotinib, with the choice between agents guided by availability, comorbidities, and toxicity management requirements.

RET Rearrangement

Recommendation 2.15.

Literature review update. This recommendation remains unchanged. Selpercatinib is a first-in-class, highly selective, and potent CNS-active RET kinase inhibitor. An analysis of patients with RET fusion-positive NSCLC (69 treatment-naïve and 247 with prior platinum-based chemotherapy) enrolled to the phase I/II, single-arm, open-label LIBRETTO-001 study of selpercatinib in patients with RET-altered cancers demonstrated that in the platinum chemotherapy pretreated population, the ORR was 61% (95% CI, 55 to 67) with 7% CR, median DoR of 28.6 months (95% CI, 20.4 to NE), and median PFS of 24.9 months. In 26 patients with measurable baseline CNS metastasis, the intracranial ORR was 85% (95% CI, 65 to 96) with 27% CR. The full safety population included 796 patients, and the safety profile of selpercatinib was consistent with previous reports.⁷⁸

The ARROW phase I/II multicohort, open-label study enrolled 233 patients between March 2017 and May 2020 (data cutoff prior to publication) with RET fusion-positive NSCLC. 92 patients had previous platinum-based chemotherapy. ORR in evaluable pretreated patients was 61% (95% CI, 50 to 71) with 6% CR. Intracranial responses were seen in all nine patients with measurable baseline intracranial metastases. Fifty-six percent of patients had an

intracranial response, and the DoR was NR, with estimates of ongoing intracranial response at 6 months of 80% and at 12 months projected to be 53%. The most common TRAEs (>20%) included elevated transaminases, neutropenia, anemia, asthenia, and constipation; grade 3 or higher TRAEs were neutropenia (18%), hypertension (11%), and anemia (10%), with no treatment-related deaths. Serious TRAEs were reported in 55 (24%) patients and most commonly (>2%) included pneumonia (10 [4%]), pneumonitis (10 [4%]), anemia (five [2%]), and neutropenia (five [2%]). Eighty-nine (38%) patients had dose reductions secondary to TRAEs.⁷⁸ Recent information from an ad hoc analysis of the ongoing phase III AcceleRET-Lung study of first-line pralsetinib compared to standard-of-care chemotherapy $\pm$ immunotherapy reported higher rates of severe and fatal infection including severe opportunistic infections in the pralsetinib arm prompting a cautionary letter to health authorities by the sponsor.⁷⁹ Before the availability of selective RET inhibitors, multi-kinase inhibitors such as vandetanib, cabozantinib, sunitinib, and lenvatinib were evaluated in patients with RET-rearranged NSCLC.¹²⁵ Objective response rates were generally below 30%, and median PFS was <6 months. Given their nonselective kinase inhibition, these agents are associated with a higher incidence of off-target toxicities. The clinical role of multikinase inhibitors in the landscape of RET fusion therapy remains uncertain.

Clinical interpretation. In the second-line setting for RET fusion-positive advanced NSCLC, selpercatinib and pralsetinib both deliver high response rates with durable benefit after prior therapy. In pretreated patients, both agents demonstrate impressive responses, with ORR approximately 61%. Selpercatinib demonstrated some CRs with strong median DoR (28.6 months) and median PFS of 24.9 months. Impressive intracranial ORR (approximately 85%) was also observed. Pralsetinib was also associated with CNS activity and 56% of patients had intracranial responses. Toxicities differed between these agents with concerns of the incidence of severe opportunistic infection with pralsetinib. In previously treated patients with RET fusion-positive NSCLC, selpercatinib or pralsetinib could be offered with choice between agents guided by availability, comorbidities, and toxicity management requirements.

NTRK Rearrangement

Recommendation 2.16.

Literature review update. This recommendation remains unchanged. Data discussing the clinical activity of entrectinib, larotrectinib, and repotrectinib in patients with NTRK fusions have been covered in detail in recommendation 1.15.

Clinical interpretation. Larotrectinib, entrectinib, and repotrectinib pivotal trials enrolled adults and pediatric patients with advanced or metastatic solid tumors harboring NTRK fusions, including NSCLC. Fifty-eight percent

in the entrectinib cohort, 77% in the larotrectinib cohort, and 29% in the repotrectinib cohort had received prior systemic therapy. Patients with treated or asymptomatic brain metastases were allowed. All three agents demonstrated high systemic and intracranial activity with durable responses in NTRK fusion-positive tumors. Toxicity was generally manageable, although safety profiles varied across agents. The panel did not offer specific recommendations on the use of repotrectinib in this setting, as available efficacy and safety data remain insufficiently mature. Findings from the phase I/II TRIDENT-1 study are currently limited to abstract form and, given the recency of the trial and lack of a full peer-reviewed publication, do not yet support definitive comparative assessment. In clinical practice, agent selection may be influenced by drug availability, comorbidities, the presence of treated or asymptomatic intracranial metastases, and anticipated toxicity management requirements. These recommendations should be interpreted in the context of evidence largely derived from single-arm, open-label phase I/II studies with relatively small patient cohorts, reflecting the rarity of NTRK fusion-positive NSCLC.

HER2 Exon 20 Mutations

Recommendation 2.17.

Literature review update. The recommendation on trastuzumab deruxtecan is unchanged, while the recommendation on zongertinib and sevabertinib is new.

The DESTINY-Lung01 trial evaluated the safety and efficacy of trastuzumab deruxtecan (T-DXd), a HER2 ADC, in patients with HER2 mutation-positive metastatic NSCLC whose disease had progressed on systemic therapy.¹²⁹ In this phase II trial involving 91 patients who had received a median of two lines of systemic therapy (including 95% platinum-based chemotherapy and approximately two-thirds a PD-(L)1 immune checkpoint inhibitor) and 36% of whom had brain metastases, T-DXd was administered at a dose of 6.4 mg/kg once every 3 weeks. The primary end point of objective response was achieved in 55% (95% CI, 44% to 65%). For the other end points, DCR was 92% (95% CI, 85% to 97%), DoR 9.3 months (95% CI, 5.7 to 14.7), median PFS 8.2 months (95% CI, 6.0 to 11.9), and median OS 17.8 months (95% CI, 13.8 to 22.1). For patients with CNS disease, median PFS was 7.1 months (95% CI, 5.5 to 9.8) while median OS was 13.8 months (95% CI, 9.8 to 20.9). TRAEs occurred in 97% of patients, with the most frequent nonhematological ones being nausea (73%), fatigue (53%), alopecia (46%), vomiting (40%), diarrhea (32%), and anorexia (30%). For these AEs, <10% were grade 3 or higher. On the other hand, among the hematological toxicities, neutropenia and anemia occurred in 35% and 33% of patients, respectively, with the frequency of grade 3 or higher TRAEs being 15% and 10%, respectively. Discontinuation of treatment due to TRAEs occurred in 25% (n = 23) of patients, while dose reduction was necessary in 34%. TRAE of special interest was ILD that

occurred in 26% and was the most common reason for treatment discontinuation (18 of 23 patients). Median onset of ILD was 20 weeks from treatment initiation and lasted for a median of 6 weeks.

The phase II DESTINY-Lung02 trial evaluated two T-DXd dosage regimens (5.4 mg/kg versus 6.4 mg/kg, 2:1 randomization) in a similar cohort of 152 patients with pretreated *HER2* mutation-positive metastatic NSCLC.¹³⁰ While the primary objective of objective response rate was comparable between the arms, 49.0% (95% CI, 39.0% to 59.1%) with 5.4 mg/kg versus 56.0% (95% CI, 41.3% to 70.0%) with 6.4 mg/kg, the lower 5.4 mg/kg dose was associated with a favorable toxicity profile. This included a marked reduction in the incidence of grade 3 or higher treatment-related AEs (39% v 58%) and treatment-emergent ILD (13% v 28%).

The phase Ia-Ib Beamion Lung-1 trial assessed the efficacy of zongertinib—an oral irreversible *HER2*-selective TKI for previously treated *HER2* mutation-positive advanced or metastatic NSCLC. Two doses, 120 mg and 240 mg once daily, were assessed, and based on an interim analysis, the 120 mg once daily dose was chosen for further evaluation. The primary end point of objective response rate was 71% (95% CI, 60% to 80%) with 120 mg once daily dose among patients with *HER2* mutation in the tyrosine kinase domain (cohort 1) compared to 30% (95% CI, 15% to 52%) for those with a nontyrosine kinase domain mutation (cohort 3). Among patients who had received prior treatment with T-DXd or any other *HER2*-directed ADC (N = 31, cohort 5), the objective response rate was 48% (95% CI, 32 to 65). For cohort 1, all subgroups (including those with prior PD-(L)1 immune check point inhibitor therapy and those with brain metastases) benefited equally in terms of objective responses and the overall DCR was 96% (95% CI, 89% to 99%). Grade 3 or higher TRAEs occurred in 17%, 25%, and 3% of patients in cohorts 1, 3, and 5, respectively. Importantly, no drug-related ILDs were reported in any of these three cohorts.⁹

The phase I/II SOHO-01 trial evaluated sevabertinib 20 mg twice daily in patients with advanced NSCLC and a *HER2* mutation, with objective response rate as the primary end point. A total of 209 patients were enrolled across three cohorts based on prior treatment exposure: Cohort D included previously treated patients who were naïve to *HER2*-directed therapies (n = 81); Cohort E included patients previously treated with a *HER2*-directed ADC (n = 55); and Cohort F included treatment-naïve patients (n = 73). The median follow-up was 13.8, 11.7, and 9.9 months in cohorts D, E, and F, respectively.

Sevabertinib demonstrated robust antitumor activity with objective response rates of 64% (95% CI, 53 to 75) in Cohort D, 38% (95% CI, 25 to 52) in Cohort E, and 71% (95% CI, 59 to 81) in Cohort F. In Cohort D, the median DoR was 9.2 months (95% CI, 6.3 to 13.5) and median PFS was 8.3 months (95% CI, 6.9 to 12.3). In Cohort E, the median DoR was 8.5 months with a median PFS of 5.5 months. For

treatment-naïve patients in Cohort F, the median DoR was 11 months, and PFS data were not yet mature at the time of reporting. The most common TRAE was diarrhea, occurring in 81%–94% of patients, with grade ≥ 3 diarrhea reported in 5%–23%. Overall, grade ≥ 3 TRAEs occurred in 31% of patients; however, treatment discontinuation due to AEs was low. Other frequently reported AEs included rash (47% to 51%) and paronychia (22% to 29%).¹⁵

Clinical interpretation. Based on the results from the DESTINY-Lung01 and DESTINY-Lung02 clinical trials, T-DXd received the initial regulatory approval for a *HER2*-directed therapy specifically for advanced or metastatic NSCLC harboring a *HER2* mutation in the previously treated setting. These studies also established 5.4 mg/kg once every 3 weeks as the preferred dose for clinical use. Importantly, both trials enrolled patients overwhelmingly with *HER2* mutations in the tyrosine kinase domain (93% in DESTINY-Lung01 and 98% in DESTINY-Lung02). While T-DXd represents a major advancement, it requires intravenous administration and is associated with significant toxicities, particularly ILD, underscoring the unmet need for an effective orally available *HER2*-targeted TKI—similar to targeted therapies already established for other oncogenic drivers in NSCLC. The ongoing phase III DESTINY-LUNG04 study (ClinicalTrials.gov identifier: [NCT05048797](https://clinicaltrials.gov/ct2/show/study/NCT05048797)) is evaluating first-line therapy with T-DXd compared to standard of care.

When comparing emerging oral TKIs in this space for patients with *HER2* kinase domain mutations, both zongertinib and sevabertinib have demonstrated promising efficacy, achieving objective radiologic response rates of approximately 60%–70%, with median DoR of 14.1 months for zongertinib and 9.2 months for sevabertinib, and median PFS of 12.4 months and 8.3 months, respectively. By comparison, T-DXd achieved an ORR of 55%, a median DoR of 9.3 months, and a median PFS of 8.2 months in similar patient populations. Additionally, zongertinib appears to have a more favorable AE profile, with a lower incidence of TRAEs compared to both T-DXd and sevabertinib. Importantly, both zongertinib and sevabertinib demonstrated activity in patients previously treated with T-DXd, although a greater number of such patients were included in the sevabertinib SOHO-01 trial compared to the Beamion Lung-1 zongertinib trial (55 v 31 patients).

Given their impressive efficacy, improved tolerability, and the convenience of oral administration, zongertinib and sevabertinib may be preferred by clinicians, patients, and caregivers over T-DXd in future treatment algorithms. Results from the ongoing phase III Beamion LUNG-2 trial (ClinicalTrials.gov identifier: [NCT06151574](https://clinicaltrials.gov/ct2/show/study/NCT06151574)) and SOHO-2 trial (ClinicalTrials.gov identifier: [NCT06452277](https://clinicaltrials.gov/ct2/show/study/NCT06452277)), which are evaluating zongertinib and sevabertinib, respectively, against standard-of-care therapies in the first-line setting for advanced or metastatic *HER2*-mutant NSCLC, are eagerly awaited. These studies will be critical in determining

whether these agents can move from their current role in pretreated disease into frontline therapy.

Recommendation 2.18.

Literature review update. This new recommendation is based on data from cohort E of the phase I/II SOHO-01 trial and cohort 5 of the phase Ia/Ib Beamion Lung-1 trial, which evaluated patients with *HER2* exon 20 mutation-positive NSCLC who had prior *HER2*-directed therapy (see Section 2.7). 41 of 55 patients in cohort E of SOHO-01 had prior T-DXd exposure, with 34% (95% CI, 20% to 51%) achieving an objective response to sevabertinib. Similarly, cohort 5 of Beamion Lung-1, which included 22 patients with prior T-DXd therapy, showed an ORR of 41% (95% CI, 23 to 61). The median DoR was 5.3 months (95% CI, 2.8 months to not evaluable) and the median PFS was 6.8 months (95% CI, 5.4 months to not evaluable).

Clinical interpretation. The SOHO-01 and Beamion Lung 1 studies evaluated a small cohort of patients with *HER2* exon 20 mutation who had received prior therapy with T-DXd. Although the absolute number of patients evaluated across both studies is small (22 in SOHO-01 and 41 in Beamion Lung 1), therapy with *HER2* TKIs achieved response rates of 30%-40%, supporting the use of this therapy in the post-ADC setting, addressing a significant unmet need in this patient population.

KRAS G12C

Recommendation 2.19.

Literature review update. This recommendation is unchanged. Sotorasib, an oral small molecule that irreversibly binds *KRAS* G12C, initially demonstrated efficacy in a specific NSCLC population in a phase II trial of 126 patients with advanced NSCLC with *KRAS* G12C mutation, 81% of whom had received prior platinum-based chemotherapy and PD-(L)1 inhibitors.¹³¹ In this nonrandomized trial, an objective response rate of 37.1% was reported, with a median DoR of 11.1 months and a median PFS of 6.8 months. This led to the randomized CodeBreaK 200 clinical trial, where 345 patients with advanced *KRAS* G12C mutation-positive NSCLC were assigned to either sotorasib (960 mg once daily, n = 171) or docetaxel (75 mg/m² once every 3 weeks, n = 174).¹³² The primary end point of BICR PFS was met: the PFS for sotorasib was 5.6 months compared to 4.5 months in the docetaxel group (HR, 0.66 [95% CI, 0.51 to 0.86], *P* = .0017). The ORR was higher in the sotorasib cohort (28.1% v 13.2%, *P* < .001); however, there was no improvement in OS (10.6 v 11.3 months, HR, 1.01 [95% CI, 0.77 to 1.33], *P* = .53). The most common AEs (occurring in >10% patients) were GI (diarrhea, nausea, decreased appetite, and transaminitis). Severe (grade 3 or worse) TRAEs were reported in 33% of patients in the sotorasib arm, most commonly diarrhea (12%) and elevated ALT (8%) or AST (5%), compared to 40% in the docetaxel group (primarily neutropenia, febrile neutropenia, and fatigue).

Adagrasib, an oral small molecule that selectively and irreversibly binds *KRAS* G12C, was studied in a phase II single-arm trial with 116 patients with *KRAS* G12C mutation-positive NSCLC, of whom 98% had previously received chemotherapy and immunotherapy. An objective response of 42.9% was reported, with a median PFS of 6.5 months and a median DoR of 8.5 months. Subsequently, the KRYSTAL-12 trial randomly assigned patients with locally advanced or metastatic *KRAS* G12C mutation-positive NSCLC, previously treated with both platinum-based chemotherapy and PD-(L)1 inhibitors, to either adagrasib (600 mg twice daily, n = 301) or docetaxel (75 mg/m² once every 3 weeks, n = 152). The trial met its primary end point of an improvement in BICR-assessed PFS (5.5 v 3.8 months; HR, 0.58 [95% CI, 0.45 to 0.76]; *P* < .0001). There was a higher ORR with adagrasib (32% v 9%, *P* < .0001). The most common AEs of any grade were diarrhea (53%), vomiting (35%), and nausea (34%). Grade 3-4 TRAEs were reported in 47% patients receiving adagrasib, the most common were elevated ALT (8%) or AST (6%) and diarrhea (5%). Dose reductions of adagrasib were required in 51% of patients compared to 24% of docetaxel patients. OS data have not been reported.^{133,134}

Clinical interpretation. Sotorasib has not demonstrated an OS advantage in the only randomized trial to date but may be offered as an option for patients with *KRAS* G12C mutation-positive NSCLC after prior platinum-based chemotherapy and PD-(L)1 inhibitors. The advantages of sotorasib over docetaxel include a significantly higher response rate, statistically significant PFS benefit, lower AE rate, and the convenience of oral therapy. Sotorasib is not recommended as a first-line option, and patients with *KRAS* G12C mutation-positive NSCLC should be offered standard first-line treatment with immune checkpoint inhibitor therapy and/or chemotherapy.

While there are no OS data yet available, adagrasib is an option for patients with *KRAS* G12C mutation-positive NSCLC after prior treatment with platinum-based chemotherapy and immune checkpoint inhibitors, based on the phase III KRYSTAL-12 trial that demonstrated significant improvements in both PFS and ORR. The oral formulation and improved quality-of-life end points favor adagrasib, but clinicians should note that dose reductions are commonly required.

NRG1 Fusion

Recommendation 2.20.

Literature review update. Recommendations for *NRG1* fusions were added in the 2025.1 version and remain unchanged for this guideline update. Zenocutuzumab is an IgG1 bispecific antibody against *HER2* and *HER3* that prevents dimerization and downstream signaling, and has emerged as a treatment option for patients with *NRG1*

fusions that have been previously treated with standard-of-care nontargeted therapy. In the phase II registrational eNRGy study,¹³⁵ patients with advanced solid tumors containing an *NRG1* fusion who had received appropriate standard therapy were enrolled and received zenocutuzumab 750 mg IV once every 2 weeks. The primary end point was investigator-assessed ORR. Among 158 response-evaluable patients, the investigator-assessed ORR was 30% (95% CI, 23 to 37) with a median DoR of 11.1 months (95% CI, 7.4 to 12.9) and a median PFS of 6.8 months (95% CI, 5.5 to 9.1). Among 93 response-evaluable patients with NSCLC, the ORR was 29% (95% CI, 20 to 39) with a median DoR of 12.7 months (95% CI, 1.8 to 29.5) and a median PFS of 6.8 months (95% CI, 5.3 to 7.5). Among 204 patients evaluated for safety, the most common TEAEs were diarrhea (29%), fatigue (21%), and nausea (20%). Grade ≥ 3 TEAEs occurred in 35%, among which anemia (5%) and elevated AST or ALT (3% each) were most common.

Clinical interpretation. Due to the low incidence of *NRG1* fusions in NSCLC, efficacy data for standard chemotherapy-immunotherapy and platinum doublet-based chemotherapy regimens are limited. The ORRs of chemotherapy-immunotherapy and platinum doublet-based chemotherapy were 0% (0/9) and 13% (2/15), respectively. Median PFS with chemotherapy-immunotherapy was 3.3 months (95% CI, 1.4 to 6.3). Retrospective evidence reports response to these regimens is suboptimal. Given this, while the eNRGy study was a single-arm phase II trial, the published efficacy data support the use of zenocutuzumab in patients with *NRG1* fusion-positive NSCLC.

Figures 1–3 provide the treatment algorithm based on these recommendations.

Special Patient Subpopulation

The data for the subpopulation, where available, are mutation-specific and not been studied across all the mutations uniformly; thus, where evidence is not mutation-specific, the recommendations can be cautiously generalized across other driver mutation classes.^{89,90,113}

Older and Frail Adults

Age alone should not preclude targeted therapy in oncogene-driven NSCLC. In ALEX, alectinib showed durable benefit over crizotinib, and the peer-reviewed trial reports include age-subgroup forest plots (<65 v ≥ 65) showing a consistent advantage with alectinib.⁵⁶ For older adults with *EGFR* mutation-positive NSCLC, a network meta-analysis comparing elderly (defined as age ≥ 65) versus nonelderly cohorts supports preserved efficacy with modern *EGFR*-TKIs,¹³⁶ and real-world data in patients ≥ 75 years on first-line osimertinib (n = 132) reported ORR approximately 75%, DCR approximately 93%, and median PFS approximately 19.4 months, reinforcing feasibility in the very old.¹³⁷ For frailty, outcomes and toxicity are better

predicted by geriatric assessment (GA) than by age alone: the GAP70+ cluster-randomized trial showed GA-guided management led to significant reduction in grade 3–5 toxicities and more medications discontinued.¹³⁸ This trial included patients with solid tumors (including lung cancer in both arms) and lymphoma. A recent ASCO guideline¹³⁹ and a position paper from the European Society for Medical Oncology and International Society of Geriatric Oncology (SIOG) both recommend integrating GA (screen $\rightarrow$ full GA) to guide treatment selection, supportive care, and dose modifications in older adults.^{139,140} For older adults with oncogene-driven NSCLC, the SIOG expert group recommends that targeted therapy should be offered, guided by a comprehensive GA and individualized to the patient's overall health status, treatment goals, and preferences.¹⁴¹ The Expert Panel agrees with these recommendations.

Pregnancy and Fertility

In a single-center retrospective analysis of 2422 women with NSCLC, among women of reproductive age (18–45 years) with NSCLC (n = 160), up to 79% harbored a targetable driver, with *ALK* and *EGFR* each present in approximately 38%; 5% of such women were diagnosed during pregnancy or the peripartum period, where *ALK* rearrangements predominated.¹⁴² In the prospective Genomics of Young Lung Cancer study, among patients with adenocarcinoma and previous genomic testing results, 77% (86/112) had a *EGFR*, *ALK*, *ROS1*, *MET*, *ERBB2*, or *RET* mutation.¹⁴³ Together, these data emphasize that—although representing a minority of lung cancer cases—women in their reproductive years carry a disproportionately high burden of targetable oncogenic drivers compared to older patient groups.

TKIs carry a biologically plausible risk of gonadotoxicity: preclinical models show germ-cell and gonadal effects, human fertility end points remain limited, and contemporary reviews recommend individualized counseling and proactive fertility-preservation discussions. In males, pathway interference (eg, c-KIT/platelet-derived growth factor receptor) may impair spermatogenesis; in females, menstrual irregularities and rare primary ovarian insufficiency have been described with vascular endothelial growth factor or *RET*-targeting TKIs—supporting sperm cryopreservation and fertility-preservation counseling before therapy.¹⁴⁴

TKIs are generally avoided during pregnancy owing to consistent embryofetal toxicity in animal studies and demonstrable placental transfer of small, lipophilic molecules; ex vivo human cotyledon work shows higher relative transfer for erlotinib than gefitinib, and multidisciplinary reviews advise postponing targeted agents until after delivery whenever possible.^{145,146} There are case reports describing successful maternal outcomes with TKIs administered during pregnancy, particularly in women with high-burden, life-threatening disease. Published examples include the use of erlotinib continued throughout

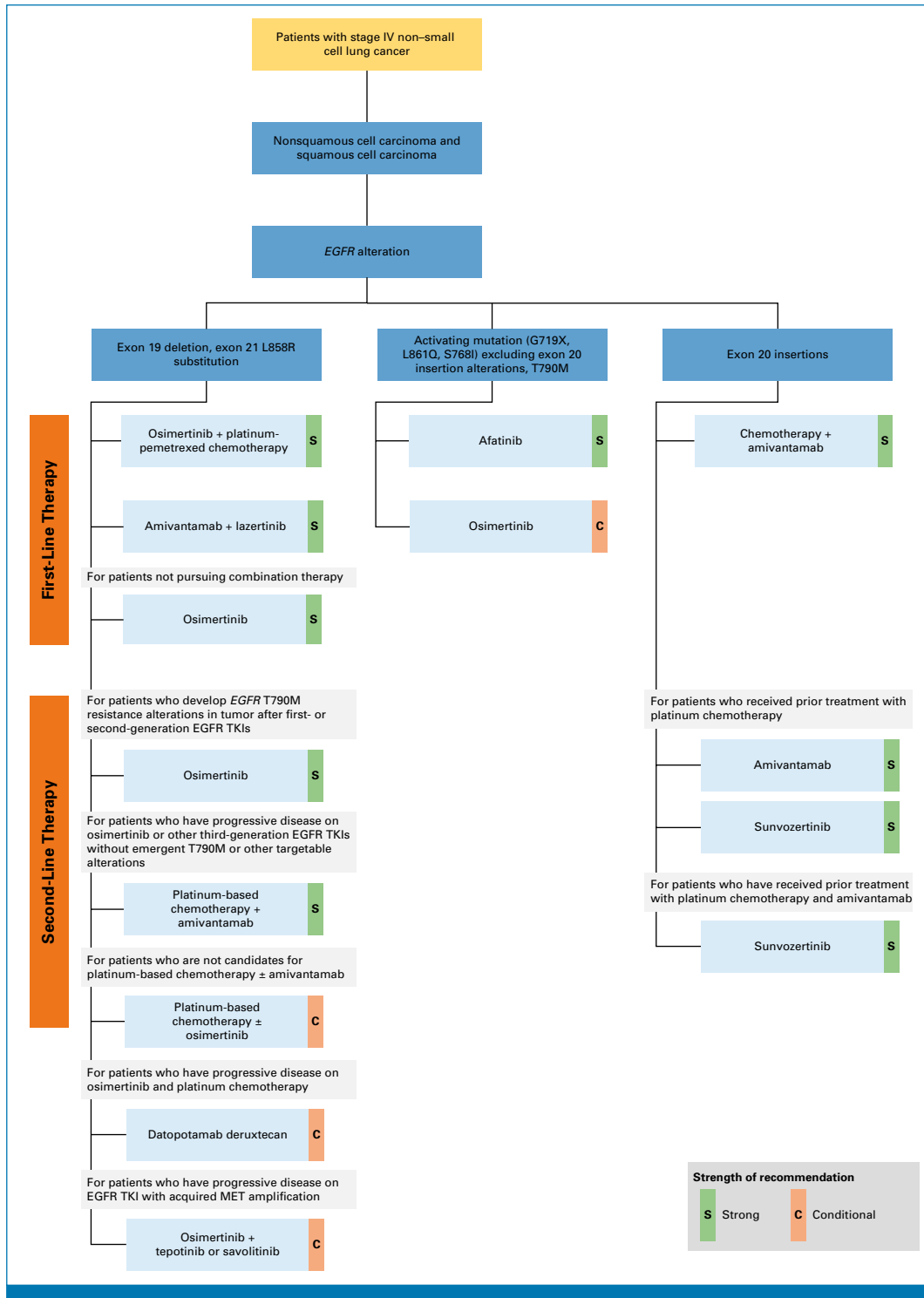


FIG 1. Treatment options for patients with stage IV non-small cell lung cancer with *EGFR* alterations. For recommendations with multiple treatment options of the same evidence quality and strength of recommendation, the decision of which agent to offer should be tailored to each patient incorporating both efficacy and toxicity. All biomarkers should be available at the time of decision making. For second-line and subsequent therapies, due to development of potentially targetable resistance mechanisms, every effort should be made to assess for presence of new mutation by tissue and/or blood NGS testing. If patients have received all targeted options, or if no targeted options are available, clinicians may offer standard therapy following the nondriver alteration guideline. EGFR, epidermal growth factor receptor; NGS, next-generation sequencing; TKI, tyrosine kinase inhibitor.

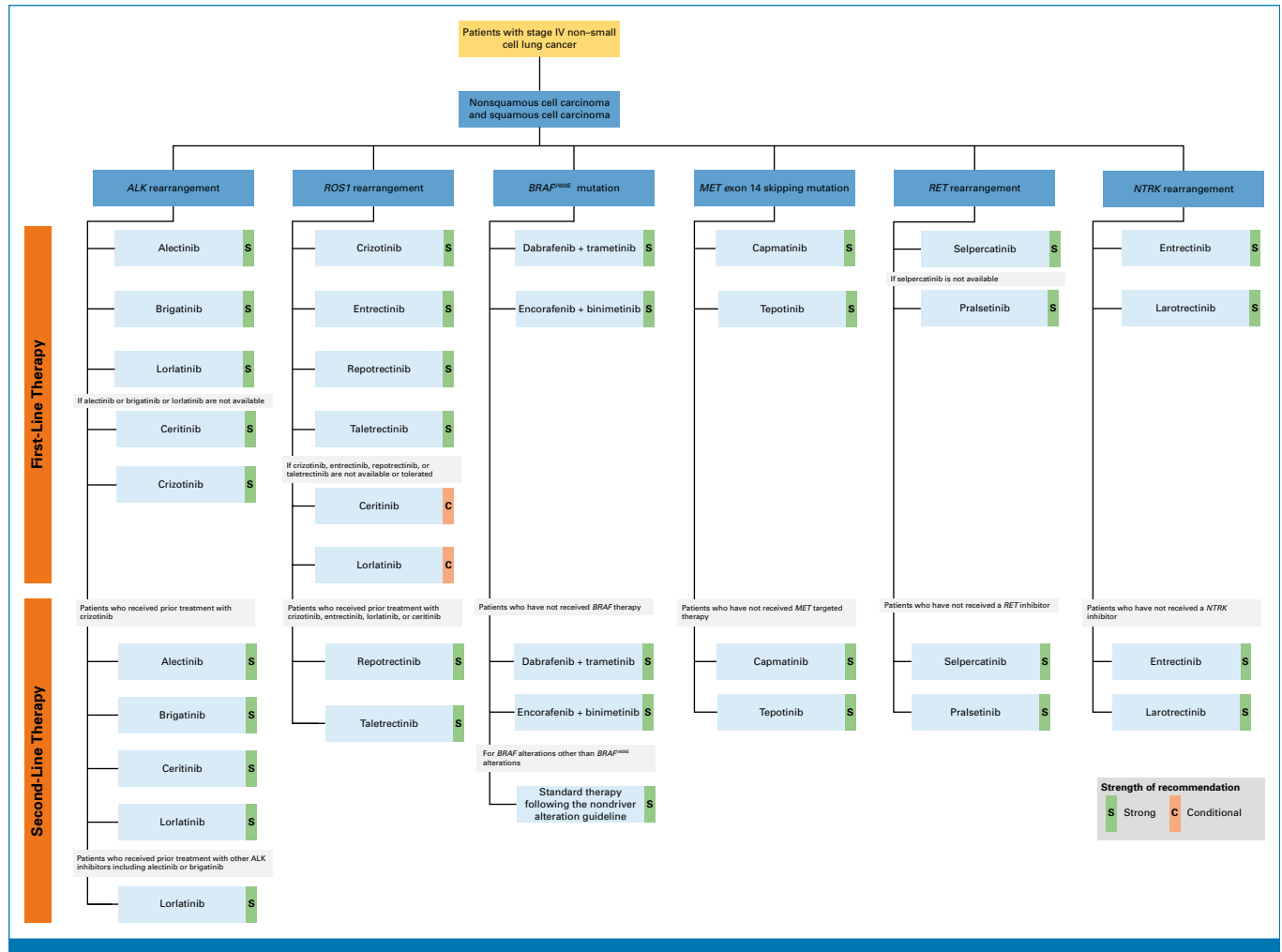


FIG 2. Treatment options for patients with stage IV non-small cell lung cancer with ALK rearrangements, *ROS1* rearrangements, *BRAF*^{V600E} mutations, MET exon skipping mutations, *RET* rearrangements, or *NTRK* rearrangements. For recommendations with multiple treatment options of the same evidence quality and strength of recommendation, the decision of which agent to offer should be tailored to each patient incorporating both efficacy and toxicity. All biomarkers should be available at the time of decision making. For second-line and subsequent therapies, due to development of potentially targetable resistance mechanisms, every effort should be made to assess for presence of new mutation by tissue and/or blood NGS testing. If patients have received all targeted options, or if no targeted options are available, clinicians may offer standard therapy following the nondriver alteration guideline. For alterations without targeted therapy options, refer to the nondriver alteration guideline, "Therapy for Stage IV Non-Small Cell Lung Cancer Without Driver Alterations: ASCO Living Guideline."¹⁷⁷ New active targeted therapies are anticipated soon. ALK, anaplastic lymphoma kinase; NGS, next-generation sequencing; NTRK, neurotrophic tropomyosin receptor kinase.

pregnancy with maternal response and peripartum holding (72 hours predelivery) followed by healthy neonatal outcomes.¹⁴⁷ Gefitinib has also been given during pregnancy, with pharmacokinetic data showing low placental transfer (approximately 20% cord blood v maternal concentration) and favorable neonatal outcomes.¹⁴⁸ Isolated case reports describe use of osimertinib and alectinib during pregnancy with maternal benefit and healthy neonatal outcomes under close monitoring; however, events such as transient oligohydramnios and intrauterine growth restriction have occurred with osimertinib, and these anecdotes do not establish safety.^{149,150}

Given the paucity of high-quality data and preclinical teratogenic signals, TKIs should generally be avoided in

pregnancy. If treatment is unavoidable for life-threatening, high-burden disease, management should be decided in a multidisciplinary discussion (medical oncology, maternal-fetal medicine, neonatology), with documented consent, preference for second or third trimester use when feasible, and selection of agents with the greatest pregnancy experience.

Please refer to the ASCO guideline on management of cancer during pregnancy¹⁵¹ and fertility preservation¹⁵² for additional information.

Oligometastatic Disease and Oligoprogression

In patients with stage IV NSCLC harboring oncogenic driver alterations who are treated with TKIs, extracranial

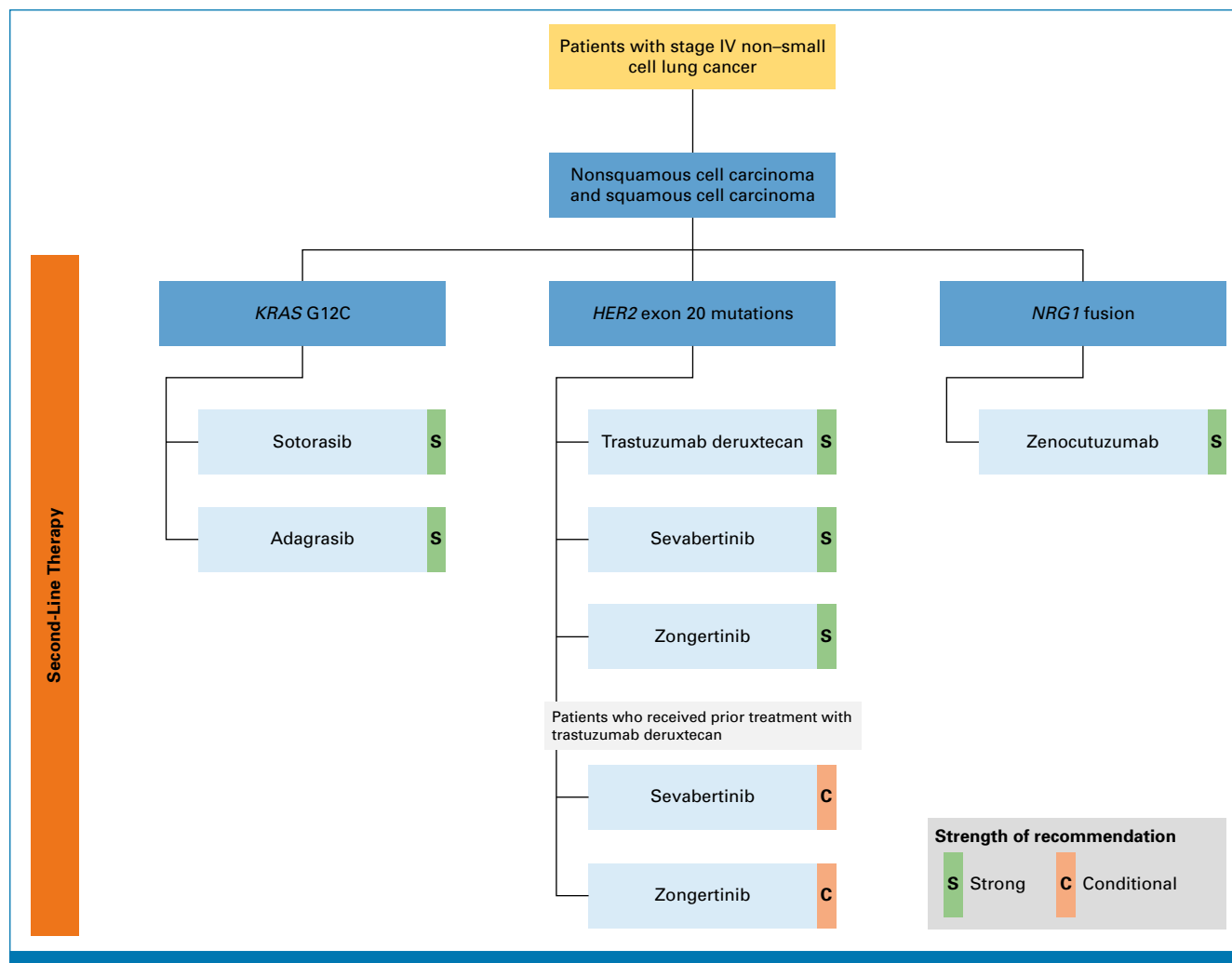


FIG 3. Second-line and subsequent treatment options for patients with stage IV non-small cell lung cancer with *KRAS* G12C alterations, *HER2* exon 20 mutations, or *NRG1* fusions. Due to development of potentially targetable resistance mechanisms, every effort should be made to assess for presence of new mutation by tissue and/or blood NGS testing. If patients have received all targeted options, or if no targeted options are available, clinicians may offer standard therapy following the nondriver alteration guideline. For alterations without targeted therapy options, refer to the nondriver alteration guideline, "Therapy for Stage IV Non-Small Cell Lung Cancer Without Driver Alterations: ASCO Living Guideline."¹⁷⁷ New active targeted therapies are anticipated soon. HER2, human epidermal growth factor receptor 2; NGS, next-generation sequencing.

oligoprogression and oligoresidual disease represent scenarios where local ablative therapy (LAT) may extend systemic control. In a Swiss multicenter cohort of first-line osimertinib, oligoprogression occurred in 77% of patients, most frequently in extracranial lung sites (62%); patients managed with LAT had the longest OS (median 60.0 months *v* 51.4 months in the non-LAT oligoprogression subset; *P* = .43).¹⁵³ In a prospective single-arm phase II trial (*n* = 61), consolidative stereotactic radiotherapy to oligoresidual disease after first-line third-generation EGFR-TKIs without interrupting therapy achieved a median PFS of 29.9 months, with a significant propensity-matched advantage versus TKI alone (HR, 0.46) and grade ≥ 3 toxicity in 6.6% of patients (mainly pneumonitis or esophagitis).¹⁵⁴ Safety and feasibility of this approach are further supported by a 2024 series evaluating extracranial stereotactic body radiation therapy in patients receiving

osimertinib or immunotherapy for induced oligometastatic states, which reported local control of 92% of the lesions treated with grade 3 pneumonitis in 2%, and delayed need for systemic therapy change—reinforcing that LAT can be delivered safely alongside TKIs.¹⁵⁵ Ongoing prospective trials (eg, NORTHSTAR, NCT03410043) are evaluating LAT integration with osimertinib to formally define benefit.¹⁵⁶

For *ALK* mutation-positive NSCLC, evidence is mainly retrospective; a recent multi-institutional cohort of alectinib-refractory patients treated for oligoprogression reported that continuation of alectinib with adjunct LAT achieved a median PFS of 8 months with minimal toxicity, supporting this strategy over immediate TKI switch.¹⁵⁷ For *ROS1* mutation-positive disease and other treatable mutations, data remain sparse, and recommendations

are extrapolated from *EGFR* and *ALK* mutation-positive populations.

Based on limited but emerging evidence, LAT may be considered for patients with extracranial oligometastatic disease while continuing the same TKI, with the aim of prolonging systemic disease control. Patient selection should be guided by a multidisciplinary team, weighing potential benefit against the risk of additive toxicity (notably pneumonitis).

LIMITATION OF RESEARCH AND FUTURE RESEARCH

Despite substantial therapeutic advances, only a subset of patients with NSCLC harbor actionable driver alterations, and among those, only a fraction derives durable clinical benefit from targeted therapies. The majority ultimately develop acquired resistance through diverse mechanisms, including secondary on-target mutations, activation of bypass signaling pathways, and phenotypic transformation. Recent advances have expanded the repertoire of potentially actionable targets to include non-G12C *KRAS* mutations, *MTAP* loss, *SMARCA4* deficiency, alterations in DNA damage repair, *STK11/LKB1*, and *NFE2L2/KEAP1* pathways. In addition, tumor-specific antigens are being leveraged for the development of ADCs and bispecific T-cell engagers. Parallel efforts are directed toward targeting resistance mechanisms through inhibition of compensatory pathways and elimination of drug-tolerant persister cell populations. Another critical unmet need is the identification of predictive biomarkers to guide selection of patients most likely to benefit from combination approaches versus single-agent targeted therapy, thereby optimizing efficacy while minimizing toxicity. Addressing these gaps requires integration of biomarker-driven adaptive trial designs, incorporation of agents with optimized CNS penetration, and longitudinal molecular profiling to capture tumor evolution. Concurrently, refining toxicity management and survivorship strategies remains essential to sustain treatment benefit and quality of life in this expanding patient population.

To enhance the global relevance of this guideline, a summary of selected drug therapies that are currently available in specific countries outside of the United States along with their key supporting clinical trials is included in Appendix [Table A3](#). This table highlights several promising therapies that may be relevant to ongoing global clinical trial participation or offer insight into future treatment directions. This table will be updated on an annual basis to serve as an informational resource for clinicians worldwide.

PATIENT AND CLINICIAN COMMUNICATION

Treatment options for stage IV NSCLC with driver alterations are evolving quickly. Next-generation targeted therapies and updated procedures (such as local

consolidative therapy for metastatic NSCLC) are helping patients live longer—sometimes for several years. There is much reason for hope. Physicians can encourage productive communication by asking how involved the patient wants to be in decision making, sharing information at the level the patient prefers, and including the patient as a full member of their care team. See the Discussion Points Between Patients and Clinicians box.

For recommendations and strategies to optimize patient-clinician communication, see Patient-Clinician Communication: American Society of Clinical Oncology Consensus Guideline.¹⁵⁸

PATIENT ACCESS CONSIDERATIONS

Not all patients have consistent access to the resources needed to benefit from ASCO's expert recommendations on best practices for prevention, screening, palliative and supportive care, and disease management of cancer. The rise of precision medicine, driven by molecular testing and targeted therapy, has significantly improved lung cancer survival over the past decade. Patients receiving targeted therapies tend to receive more rapid and durable tumor responses, better CNS disease control, and fewer treatment-related side effects than those undergoing conventional chemotherapy. Specifically, inhibitors of *EGFR*, *ALK*, *RAS*-*MAPK*, *RET*, *MET*, *BRAF*, and *NTRK/ROS1* have emerged, highlighting the importance of early genomic testing to identify the most efficacious treatment and avoid cross-toxicity with the inappropriate use of immune checkpoint inhibitors.¹⁵⁹ Despite the repeated emphasis on the importance of early biomarker testing, barriers persist between vulnerable populations and molecular testing globally. Black patients with metastatic NSCLC in the United States are significantly less likely than White patients to receive biomarker testing both before first-line therapy (29.7% v 36.6%) and at any time during the disease course (43.8% v 54.7%). These disparities persist after adjustment for baseline covariates. Social determinants of health also influence access to precision oncology; uninsured patients, those living in rural settings, and individuals with limited English proficiency are also less likely to undergo molecular testing.^{160,161}

In the United States, many patients remain unable to reap the benefits of innovative prevention and early detection programs, biomarker testing, and new cancer therapies due to challenges including lack of transportation, stable housing, adequate insurance coverage, food insecurity, health literacy, proximity to a dedicated cancer center, and cost of treatment and other services.¹⁶² Additionally, some populations experience stigma along with barriers to cancer screening, prevention, and treatment that contribute to these cancer disparities.¹⁶³

Further, geographic disparities can also impact the quality of care patients receive. Rural patients are more likely to have

DISCUSSION POINTS BETWEEN PATIENTS AND CLINICIANS

Goals of Care

Communicate treatment objectives (curative v palliative) using plain language centered around the patient's own goals and expectations.

Risks and Benefits of Therapeutic Options

Outline potential treatment options including a focus on treatment delivery, schedule, and AEs, incorporating patient-centered shared decision making.

Comorbidities

Incorporate medical comorbidities and patient-specific socioeconomic factors when determining a treatment regimen and making adjustments during treatment.

Biomarker and Genetic Testing

Explain all diagnostic testing results (imaging, molecular) and how they guide therapy, homing in on the personalized approach that is required to properly treat each patient.

Clinical Trials

Discuss the role of clinical trials and where they may (or may not) fit into the treatment course for a particular patient. Considerations include patient fitness for a study, line of therapy, and phase of trial (first-in human, phase II, phase III, etc) Common concerns, including:

Treatment Burden and Symptom Management

Proactively address symptoms and quality of life, making efforts to make adjustments (eg, dose reductions, dose delays) where appropriate. Early integration of palliative care to maximize quality of life is strongly encouraged.

Psychosocial Concerns and Available Resources

Integrate patient psychosocial and cultural beliefs into care. Utilize resources at your disposal to connect patients with appropriate resources, such as patient advocacy groups (eg, LUNGevity, Lung Cancer Research Foundation, etc) to educate and help identify a community.

Financial and Insurance Considerations and Resources

Incorporate financial considerations into decision making and utilize support networks to offset costs where possible (eg, transportation assistance, meal assistance, etc).

Survivorship Planning, Including Long-Term Follow-Up, and Monitoring

Where appropriate, integrate survivorship and long-term follow-up into routine care, including consideration of chronic side effects, symptom burden, and socioeconomic stresses.

Common Pitfalls to Avoid in Patient-Centered Communication

Information overload, overpromising or focusing on hope at the expense of reality, minimizing side effects and/or patient burdens, and making assumptions about patient goals and/or beliefs.

worse survivorship outcomes and experience higher mortality rates compared to nonrural patients. This can be attributed, in part, to a lower density of specialists and dedicated cancer centers, as only 21% of nonmetropolitan counties in the United States have one or more practicing oncologists.¹⁶⁴

On a global scale, access to biomarker testing for lung cancer remains highly uneven and, in many instances, limited to the private healthcare sector. In many low- and middle-income countries (LMICs), limited laboratory

infrastructure, high costs, and lack of reimbursement prevent routine testing for key drivers such as *EGFR*, *ALK*, or *ROS1*.¹⁶⁵ In LMICs, up to 73% of clinicians reported initiating treatment before biomarker results are available.¹⁶⁶ This lack of timely and comprehensive testing often results in missed opportunities to match patients with effective targeted therapies, ultimately contributing to poorer outcomes and widening global inequities in lung cancer care. Sustained, coordinated, and context-specific interventions are necessary to achieve global equity in biomarker testing.

In 2025, ASCO updated its guideline protocol template to include a patient access considerations section, allowing the Panel to consider patient access issues prior to starting the literature search.

Molecular testing not only improves access to targeted therapies but also to innovative therapies in the context of clinical trials. Bruno et al³² demonstrated a statistically significant association between receipt of biomarker testing and clinical trial enrollment, indicating that disparities in testing may contribute to inequities in trial participation and access to precision therapies.

Targeted therapies are often high-cost oral agents, creating a substantial financial burden. Even in high-income countries, copayments and prior insurance authorizations delay treatment initiation. In LMICs, the absence of public reimbursement or local availability often forces patients to pay out of pocket, rendering therapy unattainable. Additionally, patients may experience financial toxicity from associated expenses: travel, time off work, or supportive care, which can limit adherence even after targeted therapy is prescribed.^{167,168}

There has been a rise in policy-level advocacy with state-level legislation requiring insurance coverage for comprehensive biomarker testing and targeted therapy¹⁶⁹; however, national-level policies are necessary to avoid further geographical disparities. Programs that expand insurance coverage, negotiate drug pricing, or provide patient assistance are crucial in closing the existing gaps.¹⁷⁰ Opportunities to improve access to targeted therapy in NSCLC lie at the intersection of policy, infrastructure, drug availability, and multidisciplinary collaboration.

MULTIPLE CHRONIC CONDITIONS

Creating evidence-based recommendations to inform treatment of patients with additional chronic conditions, a situation in which the patient may have two or more such conditions—referred to as multiple chronic conditions (MCC)—is challenging. Patients with MCC are a complex and heterogeneous population, making it difficult to account for all of the possible permutations to develop specific recommendations for care. In addition, the best available evidence for treating index conditions, such as cancer, is often from clinical trials whose study selection criteria may exclude these patients in order to avoid potential interaction effects or confounding of results associated with MCC. As a result, the reliability of outcome data from these studies may be limited, thereby creating constraints for expert groups to make recommendations for care in this heterogeneous patient population.

As many patients for whom guideline recommendations apply present with MCC, any treatment plan needs to take into account the complexity and uncertainty created by the presence of MCC and highlights the importance of

RELATED ASCO GUIDELINES

- Palliative Care for Patients With Cancer¹⁷⁶ (<https://ascopubs.org/doi/10.1200/JCO.24.00542>)
- Patient-Clinician Communication¹⁵⁸ (<https://ascopubs.org/doi/10.1200/JCO.2017.75.2311>)
- Practical Assessment and Management of Vulnerabilities in Older Patients Receiving Systemic Cancer Therapy¹³⁹ (<https://ascopubs.org/doi/10.1200/JCO.23.00933>)
- Fertility Preservation in People With Cancer¹⁵² (<https://ascopubs.org/doi/10.1200/JCO-24-02782>)
- Management of Cancer During Pregnancy¹⁵¹ (<https://ascopubs.org/doi/10.1200/JCO-25-02115>)
- Therapy for Stage IV Non-Small Cell Lung Cancer Without Driver Alterations¹⁷⁷ (<https://ascopubs.org/doi/10.1200/JCO-25-02825>)

shared decision making regarding guideline use and implementation. A patient's chronic condition (eg, diabetes or heart failure) may be of concern when choosing a treatment regimen due to a drug's toxicity profile. Similarly, their medications may be a concern due to drug-drug interactions. The patient's other specialists and oncology pharmacists often must be consulted in treatment planning, particularly with targeted therapies.

In light of these considerations, practice guidelines should provide information on how to apply the recommendations for patients with MCC, perhaps as a qualifying statement for recommended care. This may mean that some or all of the recommended care options are modified or not applied, as determined by best practice in consideration of any MCC.

COST IMPLICATIONS

Increasingly, individuals with cancer are required to pay a larger proportion of their treatment costs through deductibles and coinsurance.^{171,172} Higher patient out-of-pocket costs have been shown to be a barrier to initiating and adhering to recommended cancer treatments.^{173,174}

Discussion of cost can be an important part of shared decision making.¹⁷⁵ Clinicians should discuss with patients the use of less expensive alternatives when it is practical and feasible for treatment of the patient's disease and there are two or more treatment options that are comparable in terms of benefits and harms.¹⁷⁵

GUIDELINE IMPLEMENTATION

ASCO guidelines are developed for implementation across health settings. Each ASCO guideline includes a community oncologist member on the panel. The additional role of this

community oncologist member on the guideline panel is to assess the suitability of the recommendations for implementation in the community setting, but also to identify any other barrier to implementation a reader should be aware of. Barriers to implementation include the need to increase awareness of the guideline recommendations among frontline practitioners and survivors of cancer and caregivers, and also to provide adequate services in the face of limited resources. The guideline recommendations table and accompanying tools (available at www.asco.org/thoracic-cancer-guidelines) were designed to facilitate implementation of recommendations. This guideline will be distributed widely. ASCO guidelines are posted on the ASCO website and most often published in the *Journal of Clinical Oncology*.

ASCO believes that cancer clinical trials are vital to inform medical decisions and improve cancer care, and that all patients should have the opportunity to participate.

ADDITIONAL RESOURCES

For current information, including selected updates, supplements, and clinical tools and resources, visit www.asco.org/thoracic-cancer-guidelines. The Data Supplement for this guideline includes the search strategy string and archived recommendations. Guideline recommendations and algorithms are also available in the free

ASCO Guidelines app (available for download in the [Apple App Store](https://apps.apple.com/us/app/ascoguidelines/id1450865425) and [Google Play Store](https://play.google.com/store/apps/details?id=com.asco.guidelines)). Listen to key recommendations and insights from panel members on the [ASCO Guidelines podcast](https://www.asco.org/podcast). The Methodology Manual (available at www.asco.org/guideline-methodology) provides additional information about the methods used to develop this guideline. Patient information is available at www.cancer.org.

ASCO welcomes your comments on this guideline, including implementation challenges, new evidence, and how this guideline impacts you. To provide feedback, contact us at guidelines@asco.org. Comments may be incorporated into a future guideline update. To submit new evidence or suggest a topic for guideline development, complete the form available at www.asco.org/guidelines.

INCLUSIVE LANGUAGE

ASCO is committed to promoting the health and well-being of all patients. ASCO guidelines are intended to apply to, and be discussed clearly and compassionately with, all patients. For this reason, guideline authors use appropriately inclusive language. In instances in which the guideline draws upon data based on research in a specified population (eg, studies regarding women with ovarian cancer), the guideline authors describe the characteristics and results of the research as reported.

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EDITOR'S NOTE

This American Society of Clinical Oncology (ASCO) Clinical Practice Guideline provides recommendations, with comprehensive review and analyses of the relevant literature for each recommendation. Additional information, including a supplement with additional evidence tables, clinical tools and resources, and links to patient information at www.cancer.org, is available at www.asco.org/thoracic-cancer-guidelines.

EQUAL CONTRIBUTION

S.P., N.B.L., J.R., and L.B. were Expert Panel Co-Chairs.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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Conception and design: All authors

Collection and assembly of data: All authors

Data analysis and interpretation: All authors

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Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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TABLE A1. Stage IV Non–Small Cell Lung Cancer Living Guideline Expert Panel Membership

Name	Affiliation	Role or Area of Expertise
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Sonam Puri, MD	Moffitt Cancer Center, Tampa, FL	Medical Oncology
Joshua E. Reuss, MD	Georgetown University, Washington, DC	Medical Oncology
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Toby Campbell, MD	University of Wisconsin, Madison, WI	Palliative Oncology
Greg Durm, MD	Indiana University Health, Indianapolis, IN	Medical Oncology
Jill Feldman, MA	EGFR Resisters patient advocacy group, Deerfield, IL	Patient Research Advocate
Narjust Florez, MD	Dana-Farber Cancer Institute, Harvard Medical School, Boston, MA	Medical Oncology
Janet Freeman-Daily, MSc, Engr	The ROS1ders, Seattle, WA	Patient Research Advocate
Naoki Furuya, MD, PhD	St Marianna University School of Medicine, Kawasaki, Japan	Medical Oncology
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Sara Kuruvilla, MD (Ontario Health representative)	London Health Sciences Centre, London, ON, Canada	Medical Oncology
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Logan Roof, MD	Ohio State University, Columbus, OH	Medical Oncology
Erin L. Schenk, MD, PhD	University of Colorado Anschutz Medical Center, Aurora, CO	Medical Oncology
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Navneet Singh, MD, DM	Postgraduate Institute of Medical Education and Research, Chandigarh, India	Medical Oncology
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Ana I. Velazquez, MD, MSc	University of California, San Francisco, CA	Medical Oncology
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Nofisat Ismaila, MD	American Society of Clinical Oncology (ASCO), Alexandria, VA	ASCO Practice Guidelines Staff (Health Research Methods)

TABLE A2. Evidence Quality and Recommendation Rating Definitions

Term	Definition
Evidence quality	
High	We are very confident that the true effect lies close to that of the estimate of the effect.
Moderate	We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low	Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect.
Very low	We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.
Strength of recommendation	
Strong	In recommendations for an intervention, the desirable effects of an intervention outweigh its undesirable effects. In recommendations against an intervention, the undesirable effects of an intervention outweigh its desirable effects. All or almost all informed people would make the recommended choice for or against an intervention.
Conditional	In recommendations for an intervention, the desirable effects probably outweigh the undesirable effects, but appreciable uncertainty exists. In recommendations against an intervention, the undesirable effects probably outweigh the desirable effects, but appreciable uncertainty exists. Most informed people would choose the recommended course of action, but a substantial number would not.

NOTE. GRADE Handbook.¹⁷⁸**TABLE A3.** Lung Cancer Drugs Available Outside the United States

Drug (generic)	Class/Mechanism	Indication	Country	Trial Name/Reference
Firmonertinib	EGFR TKI	NSCLC, ex20ins	China	FURVENT
Anlotinib	VEGFR/EGFR TKI	Third/subsequent line	China	ALTER (PMID: 30098152)
Icotinib	EGFR TKI	First-line treatment	China	CHALLENGE (PMID: 35862035)
Iruplinalkib	ALK	First-line treatment	China	INSPIRE (PMID: 38280448)
Envonalkib	ALK	First-line treatment	China	PMID: 37574511
Befotertinib	EGFR T790M	First-/second-line treatment	China	PMID: 35724798
Sacituzumab tirumotecan	EGFR TKI (post EGFR TKI)	Second-/third-line treatment	China	NCT07193160
Savolitinib	MET exon 14	First-/second-line treatment	China	NCT02897479

NOTE. The following therapies included are based on trials meeting the evidence requirement for inclusion in this guideline and this list is for informational purposes only. These therapies are available in specific jurisdictions outside of the United States. Readers should consult local regulatory authorities/product monographs for the most up-to-date status.

Abbreviations: ALK, anaplastic lymphoma kinase; EGFR, epidermal growth factor receptor; NSCLC, non–small cell lung cancer; TKI, tyrosine kinase inhibitor; VEGFR, vascular endothelial growth factor.